

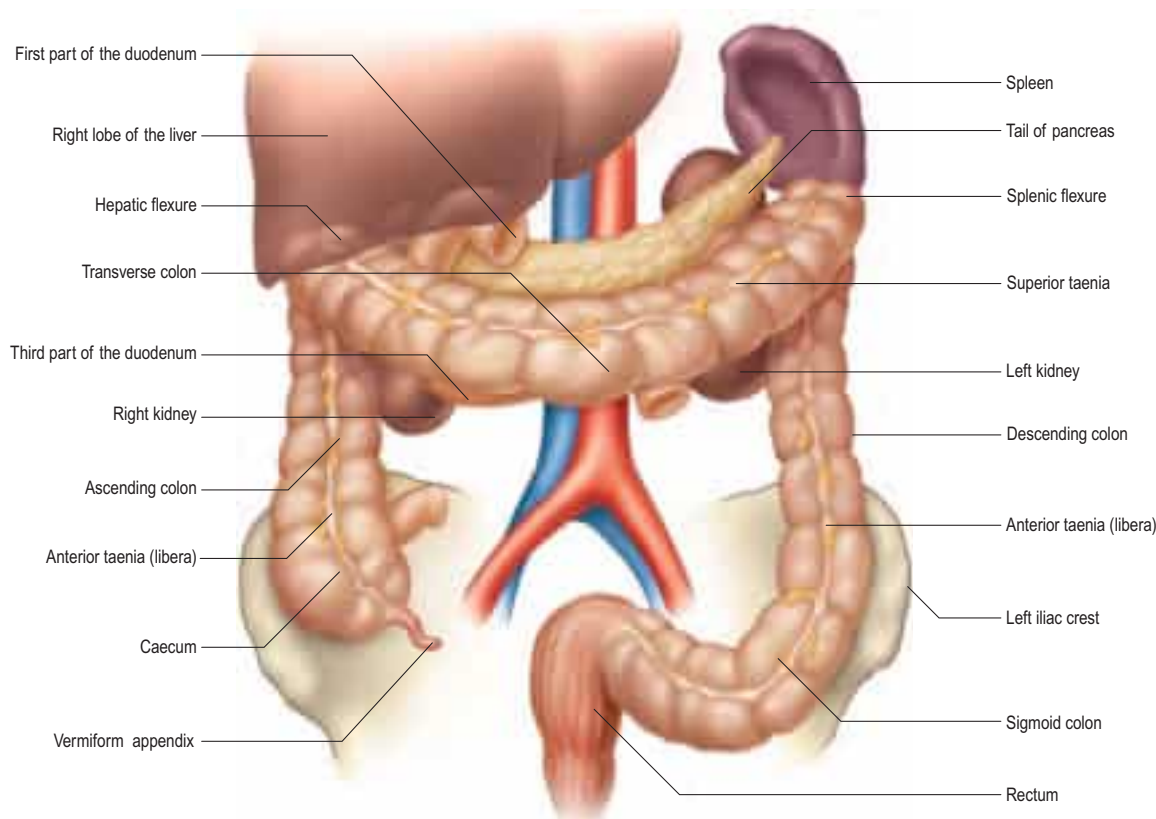
OVERVIEW

The large intestine extends from the ileocaecal junction to the anus. It begins as the caecum and vermiform appendix, which are usually located in the right iliac fossa. The ascending or right colon passes upwards in the right flank to the right hypochondrium, where it bends to the left to form the hepatic flexure (right colic flexure) and become the transverse colon. This loops across the abdomen with an antero-inferior convexity until it reaches the left hypochondrium, where it curves inferiorly to form the splenic flexure (left colic flexure). From here, it descends in the left flank as the descending or left colon before continuing as the sigmoid colon in the left iliac region. The sigmoid colon descends into the true pelvis and becomes the rectum anterior to the third sacral vertebra. The rectum transitions to the anal canal at the level of the pelvic floor (Figs 66.1–66.2). The large intestine thus runs from the ileocaecal (ileocolic) junction to the anal verge. It is formed from the distal midgut, all of the hindgut, and the proctodeum (Ch. 72 and Figs 72.4 and 72.7E). Knowledge of the development of the large intestine not only helps to explain its anatomy, relations (including peritoneal attachments) and neurovascular supply (Ch. 60), but is also fundamental to understanding congenital large bowel disorders such as anorectal malformations and malrotation of the gut.

In the adult, the large intestine is approximately 1–1.5 m long *in vivo*, although there is considerable individual variation. The large intestine differs from the small intestine in several ways: it has a greater calibre; for much of its course it is more fixed in position; the outer longitudinal muscle layer of the colon is concentrated into three longitudinal bands, taeniae coli; small fatty projections, appendices epiploicae, are scattered over its free surface (although these tend to be

absent from the caecum, vermiform appendix and rectum); and the colonic wall is puckered into sacculations (haustrations), visible on plain radiographs as incomplete septations arising from the bowel wall (see Fig. 66.2; Fig. 66.3). Its calibre is greatest near the caecum, gradually diminishes towards the sigmoid colon, and then increases again in the rectum, where the lower third is dilated to form the rectal ampulla. The mean internal diameter of the large intestine is 4.8 cm (Helander and Fändriks 2014).

During development, the large intestine is temporarily suspended by a midline dorsal mesentery. However, after rotation of the gut *in utero*, large portions of it come to lie adherent to the retroperitoneum, while other segments remain suspended by a mesentery within the peritoneal cavity. The mesenteries of the colon consist of two layers of visceral peritoneum enclosing adipose and connective tissues and surrounding vessels, nerves and lymphatics that run forward from retroperitoneal structures. Where colonic mesenteries lie in contact with the retroperitoneum, the potential space between the retroperitoneum and mesentery, referred to as the 'subperitoneal space', forms an avascular plane during surgical dissection and allows the tracking of fluid, blood and disease (Oliphant et al 1996, Coffey 2013) (Ch. 62). The caecum may be firmly adherent to the retroperitoneum but is frequently suspended by a short mesentery, especially in infants. The ascending colon is usually adherent to the retroperitoneum, while the transverse colon is suspended by a mesocolon and is freely mobile within the upper abdomen. The hepatic and splenic flexures may have a short mesentery. The descending colon is adherent to the retroperitoneum, usually down to the level of the left iliac crest, and the sigmoid colon has a mesentery of variable length. The sigmoid mesentery shortens as it approaches the pelvis and practically disappears at the level of the rectosigmoid junction. The rectum is a retroperitoneal structure.



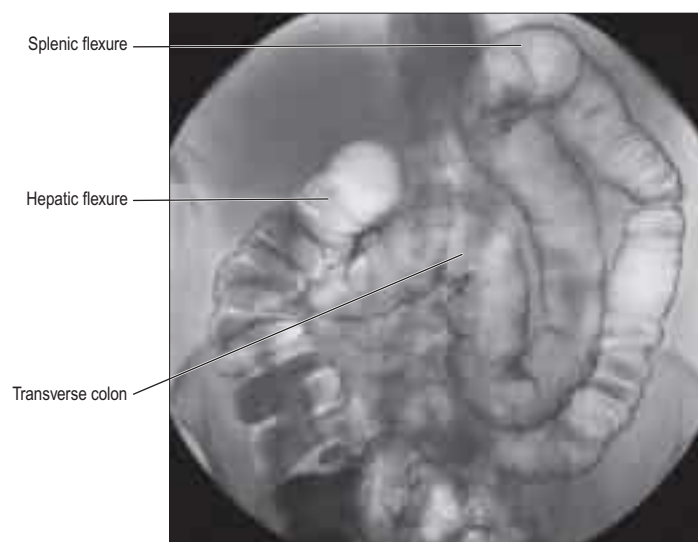


Fig. 66.2 The appearance of the colon on double contrast barium enema examination, demonstrating the transverse colon and hepatic and splenic flexures.

The posterior attachments of the small bowel mesentery and transverse mesocolon result in the peritoneal cavity being divisible into named compartments (**Ch. 62**). The supracolic and infracolic compartments are separated by the attachment of the transverse mesocolon to the posterior abdominal wall. Anteriorly, the transverse colon and transverse mesocolon are adherent to the posterior surface of the greater omentum. Lifting the greater omentum upwards exposes the infracolic compartment. This is divided by the root of the small bowel mesentery that descends obliquely from left to right; to the right lies the right infracolic compartment, its apex the ileocaecal junction, and to the left, the left infracolic compartment, bordered laterally by the descending colon and continuous below with the pelvic cavity.

The microstructure of the large bowel corresponds to the general pattern of the gut wall, having a mucous membrane consisting of epithelium, lamina propria and muscularis mucosa, surrounded by an inner circular and outer longitudinal layer of muscle and a serosa (see **Figs 66.3, 66.51**). The mucosa also contains scattered neuroendocrine cells derived from the amine precursor uptake and decarboxylation (APUD) cell lineage; these cells produce amines and/or peptides, which act as hormones or neurotransmitters. Unlike the small intestine, the mucosa of the large intestine lacks villi and the glands (crypts) contain a high proportion of goblet (mucin-secreting) cells. In the appendix, the glands are sparse and numerous lymphoid follicles are found in the mucosa and submucosa. Although it is not a vital organ, the primary function of the colon is transmission and elimination of intestinal contents, but it also has important absorptive and secretory roles. The anal sphincter is responsible for both continence and evacuation. The large intestine has its own intrinsic (enteric) nerve supply, the activity of which is modulated by extrinsic innervation.

EXTERNAL APPEARANCE

The haustrations of the colon are frequently absent in the caecum and often relatively sparse in the ascending and proximal transverse colon. They become more pronounced beyond the middle of the transverse colon. The sigmoid colon often has marked sacculations. Appendices epiploicae are small, fat-filled pouches of peritoneum that project from the external surface of the colon and are supplied by blood vessels that perforate the bowel wall. There are few, if any, appendices epiploicae on the serosal surface of the caecum, and only a limited number scattered along the ascending colon. They are more common over the distal colon and particularly numerous on the surface of the sigmoid colon, where they can be large in the obese individual. The rectum has no appendices epiploicae.

The three taeniae coli are located in fairly constant positions beneath the serosal surface of the colon, except in the transverse colon. They are found on the anti-mesenteric aspect of the colon directly opposite the mesentery (taenia libera), posterolaterally (taenia omentalis) and posteromedially (taenia mesocolica) midway between the taenia libera and the mesentery (see **Fig. 66.3**). In the ascending and descending colon, the posterolateral taenia is often obscured from view by the peritoneal

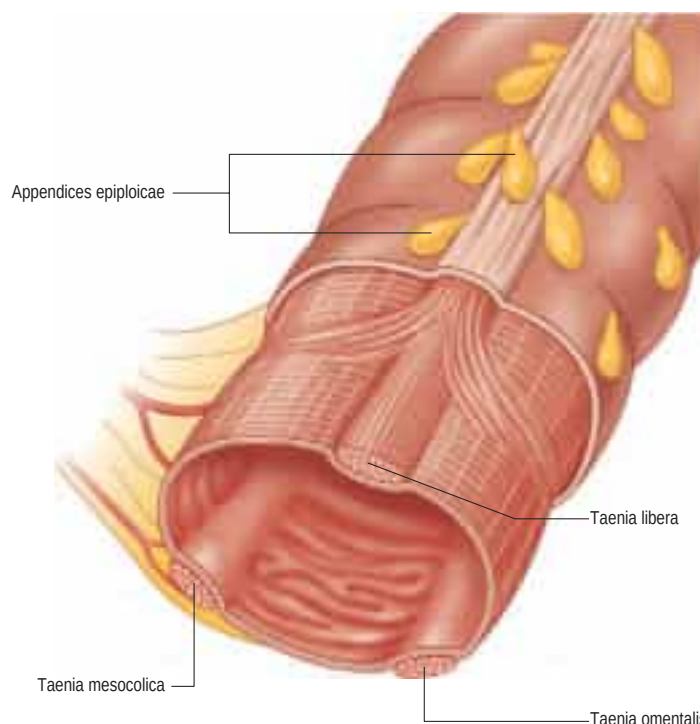


Fig. 66.3 Layers of the colonic wall.

reflection on to the colonic wall. In the transverse colon, the taeniae are rotated as a consequence of the dependent position of this segment of colon and so anterior becomes inferior, posteromedial becomes posterior, and posterolateral becomes superior. The width of the taeniae coli remains fairly constant throughout the colon but they broaden out in the distal part of the sigmoid colon and gradually merge to form a complete longitudinal muscle layer around the rectum, which therefore has no external sacculations.

INTERNAL APPEARANCE

Colonic haustrations represent sites where the mucosa and submucosa of the colonic wall is infolded; these folds partially span the bowel lumen but never form a complete, circumferential ring. The pattern of haustrations and appearance of the colonic mucosa help the clinician appreciate the level reached during flexible endoscopic examinations of the colon (**Silverstein and Tytgat 1991**). In the caecum, the three longitudinal taeniae coli converge to form a characteristic 'trefoil' pattern on the caecal wall (**Fig. 66.4**). The lower pole of the caecum is usually devoid of haustrations, although a spiral mucosal pattern is often seen in the region of the orifice of the appendix (**Fig. 66.5**). The upper caecum and ascending colon possess shallow but deep haustrations, which may extend across one-third of the lumen (**Fig. 66.6**). In the transverse colon, the haustrations often confer a triangular appearance to the cross-section of the lumen when viewed along its axis at colonoscopy (**Fig. 66.7**). The haustrations of the descending and sigmoid colon tend to be thicker and shorter, producing a more circular cross-section to the lumen (**Figs 66.8–66.9**). The wall of the colon is thinnest in the region of the caecum and ascending colon, where it is most at risk of perforation during therapeutic endoscopic procedures. The overall luminal diameter is often smallest in the descending colon. The tortuosity of the sigmoid colon means that shorter lengths of colon are visible during endoscopy than elsewhere in the colon. The haustrations of the rectum usually form consistent and recognizable transverse folds and the submucosal vessels tend to be more pronounced than in the colon (**Fig. 66.10**). Distinct veins are usually visible during endoscopy, and are most marked above the anorectal junction.

RADIOGRAPHIC APPEARANCES

Cross-sectional imaging of the colon can be performed with computed tomography (CT) and magnetic resonance imaging (MRI). On axial imaging, the colon may be filled with particulate faeces and air (**Fig. 66.11**). The wall in normal individuals is thin. The caecum and ascending colon often contain faecal residue and are easily identified in the



Fig. 66.4 The endoscopic appearance of the caecum. The characteristic trefoil appearance of the confluence of the three taeniae is usually obvious.



Fig. 66.5 The endoscopic appearance of the orifice of the appendix, seen as a slit-like depression near the centre of this view of the caecal pole. The orifice varies from a small depression to an obvious luminal structure. The ileocaecal junction is just visible in the top left corner. (Courtesy of Dr Michael Schultz.)

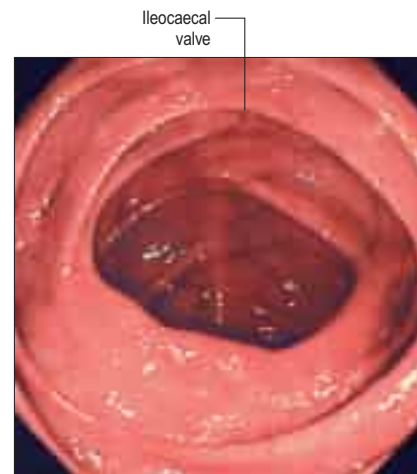


Fig. 66.6 The endoscopic appearance of the ascending colon.



Fig. 66.7 The endoscopic appearance of the transverse colon. Note the characteristic triangular appearance of the haustrations when viewed collectively (see also Fig. 66.11C).



Fig. 66.9 The endoscopic appearance of the sigmoid colon. Multiple large mucosal folds are characteristic.



Fig. 66.8 The endoscopic appearance of the descending colon. The lumen tends to look rather more featureless than the more proximal colon.



Fig. 66.10 The endoscopic appearance of the rectum. Note the large transverse folds, with little else in the way of mucosal folds, characterizing the rectum. Prominent submucosal vessels are often seen, particularly in the lower third.

retroperitoneum on the right. The transverse colon may contain faeces or gas and lies in a variable position, suspended by its mesentery. The descending colon lies in the retroperitoneum on the left and often contains little faecal residue. The volume data sets produced by modern multislice CT can now produce virtual colonoscopic mucosal images of the distended and cleaned colon, and surface-rendered images of the internal surface of the bowel (Fenlon 2002).

VASCULAR SUPPLY AND LYMPHATIC DRAINAGE

Arteries

The arterial supply of the large intestine is derived from both the superior and the inferior mesenteric arteries (Fig. 66.12) (Jackson 1999). The caecum, appendix, ascending colon and proximal two-thirds of the transverse colon (derived from the midgut) are supplied from ileocolic, right colic and middle colic branches of the superior mesenteric artery. The distal third of the transverse colon, descending and sigmoid colon, rectum and upper anal canal (hindgut derivatives) are supplied predominantly from the inferior mesenteric artery via the left colic, sigmoid

and superior rectal arteries, with a small contribution from branches of the internal iliac artery (the middle and inferior rectal arteries).

The arteries of the midgut and hindgut contribute to an anastomotic vessel, the marginal artery of Drummond, which runs in the mesentery along the inner margin of the colon and gives off short terminal branches to the bowel wall. These divide into vasa brevia, which pass directly through the muscularis externa of the colonic wall, and vasa longa, which travel through the subserosa for a short distance before running through the circular muscle, giving off branches to the appendices epiploicae (Fig. 66.13). The marginal artery is formed by the main branches and arcades arising from the ileocolic, right colic, middle colic and left colic arteries. It is most apparent in the ascending, transverse and descending colons (Fig. 66.14) and poorly developed in the sigmoid colon. The marginal artery in the region of the splenic flexure may be absent or of insufficient calibre to be of functional significance. Nevertheless, it may dilate considerably when one of the main visceral arteries is compromised since it then provides a collateral arterial supply to the colon. The most susceptible part of the chain of anastomosing vessels is at the junction of midgut and hindgut in the distal transverse colon near the splenic flexure. However, the arterial supply to this

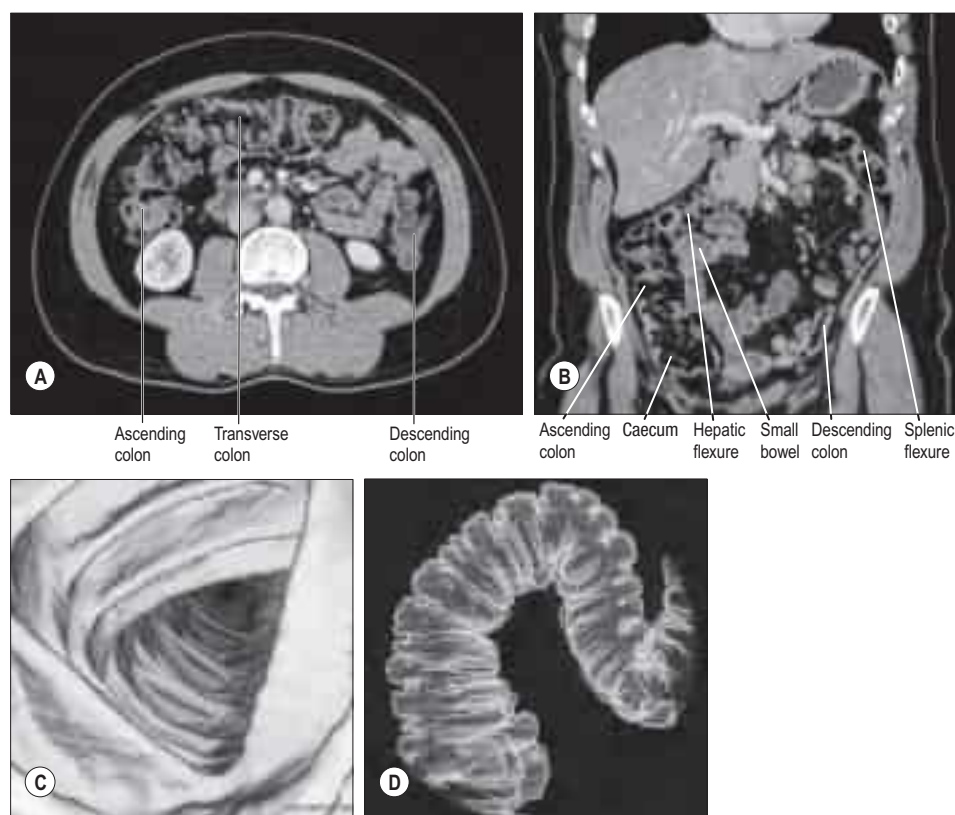


Fig. 66.11 The appearance of the colon on multislice computed tomographic examination. **A**, Axial CT. **B**, Coronal reformat showing normal calibre and distribution of the abdominal colon (ascending, transverse and descending). **C**, Volume-rendering of the colonic wall using the axial data set to produce virtual colonoscopic views, showing the triangular lumen of the transverse colon. **D**, Volume-rendering of the air-filled colon using the axial data set to give an image similar to a double contrast barium enema, demonstrating haustrations. (A–B, By courtesy of Dr Louise Moore, Chelsea and Westminster Hospital. C–D, By courtesy of GE Worldwide Medical Systems.)

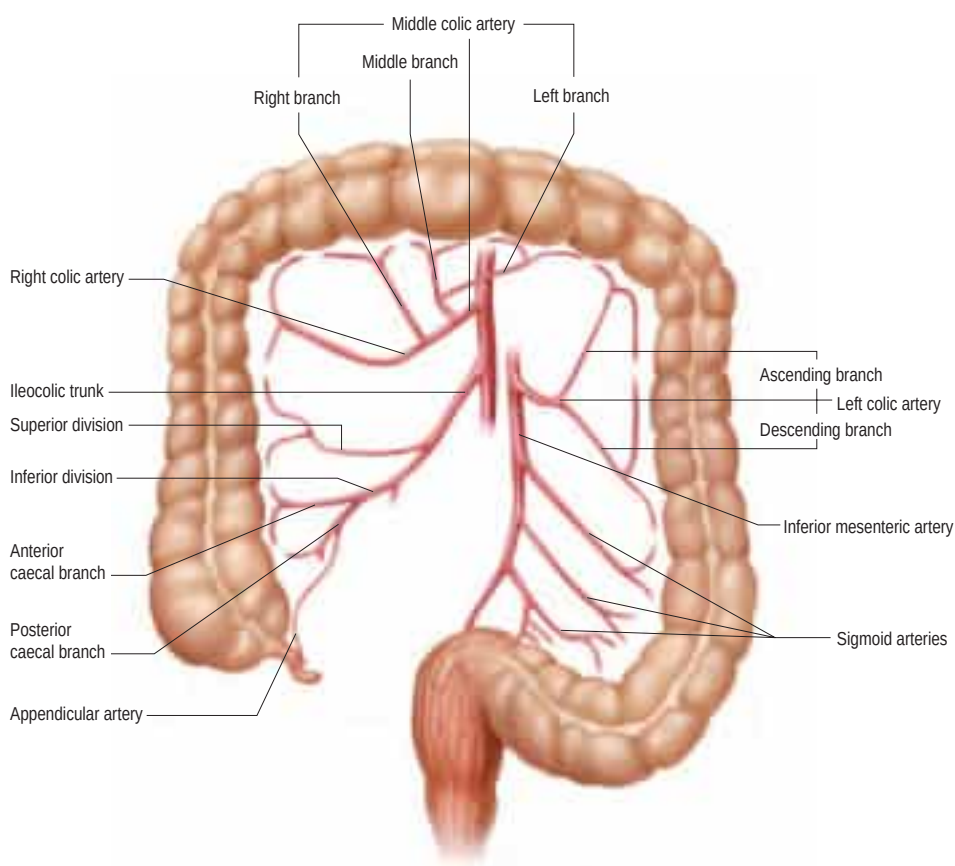


Fig. 66.12 The main branches of the superior and inferior mesenteric arteries.

region may be augmented by an inner arterial arc (of Riolan), which runs a meandering course in the colonic mesentery between the main trunk of the middle colic artery and the ascending branch of the left colic artery (Fisher and Fry 1987, Gourley and Gering 2005). When present, this vessel is usually only prominent when there is occlusion of the superior or inferior mesenteric artery.

Colonic vascular occlusion

The marginal artery of the colon may become massively dilated when there is chronic stenosis or occlusion of the superior or inferior mesenteric artery. The latter may occur, for example, in association with

an abdominal aortic aneurysm, when the descending colon remains viable because the marginal artery continues to receive an adequate blood supply from the left branch of the middle colic artery and the sigmoid colon gains its supply from the middle and inferior rectal arteries via the superior rectal and sigmoid arteries. When colonic ischaemia does occur, it is usually maximal in the region of the splenic flexure and proximal descending colon because this segment is furthest from the collateral arterial supplies. Occlusion of the common iliac arteries may also result in dilation of the marginal and inferior mesenteric arteries, which become an important collateral supply to the lower limbs via dilated middle rectal arteries arising from the internal iliac artery.

Veins

The venous drainage of the large intestine is primarily into the portal vein via the superior mesenteric and inferior mesenteric veins, although some drainage from the rectum occurs via middle rectal veins into the internal iliac veins, and inferior rectal veins into the internal pudendal veins. Those parts of the colon derived from the midgut (caecum, appendix, ascending colon and proximal two-thirds of the transverse colon) drain into colic branches of the superior mesenteric vein, while hindgut derivatives (distal third of the transverse colon, descending and sigmoid colon, rectum and upper anal canal) drain into the inferior mesenteric vein (Fig. 66.15).

Lymphatic drainage

Lymph drainage from the large intestine follows the course of the arteries. Thus, lymphatic vessels of the caecum, ascending and proximal transverse colon drain ultimately into lymph nodes related to the superior mesenteric artery, while those of the distal transverse colon,

descending colon, sigmoid colon and rectum drain into nodes following the course of the inferior mesenteric artery (Fig. 66.16). In cases where the distal transverse colon or splenic flexure is predominantly supplied by vessels from the middle colic artery, the lymphatic drainage of this area may be predominantly to superior mesenteric nodes.

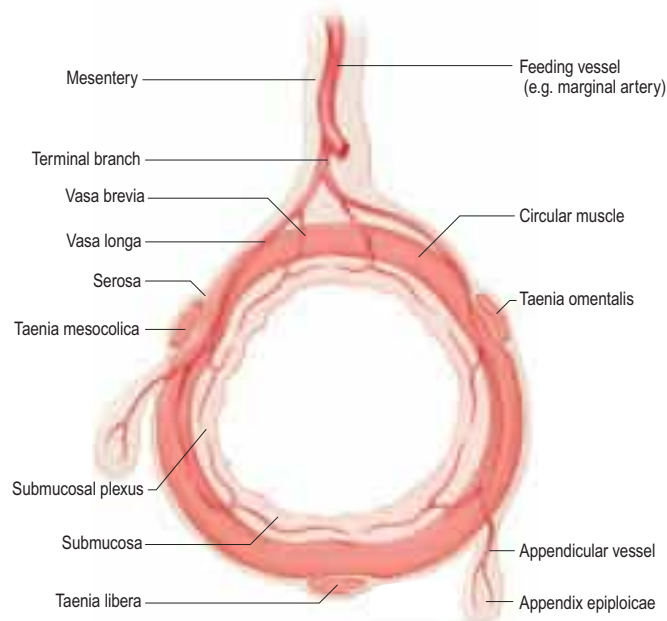


Fig. 66.13 The typical pericolic arrangement of the arterial vasculature.

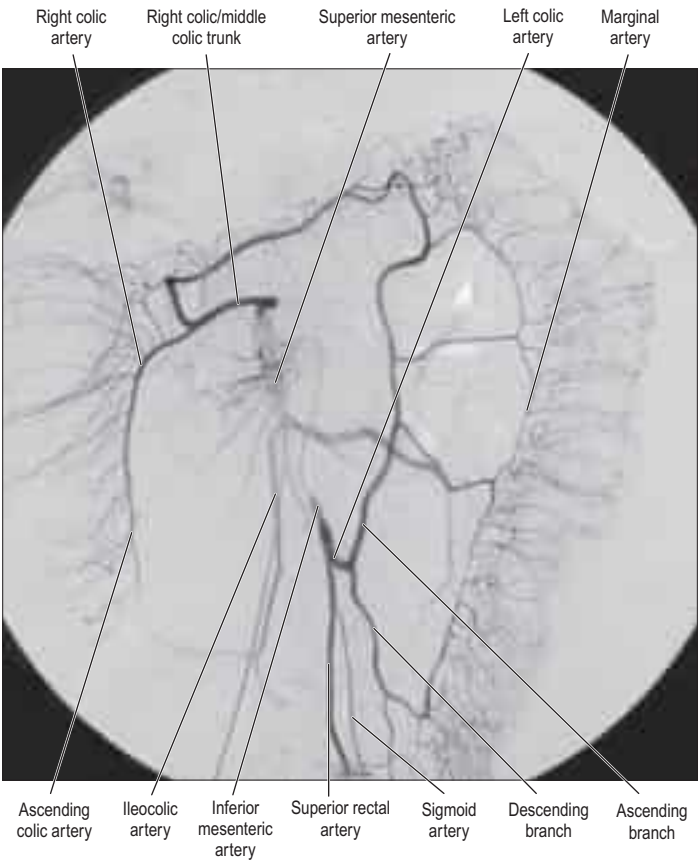


Fig. 66.14 The marginal artery running parallel to the colon and providing an anastomosis between the branches of the superior mesenteric artery supplying the right side of the colon and the branches of the inferior mesenteric artery supplying the left side of the colon (digital subtraction arteriogram). (Courtesy of Dr J Jackson, Hammersmith Hospital, London.)

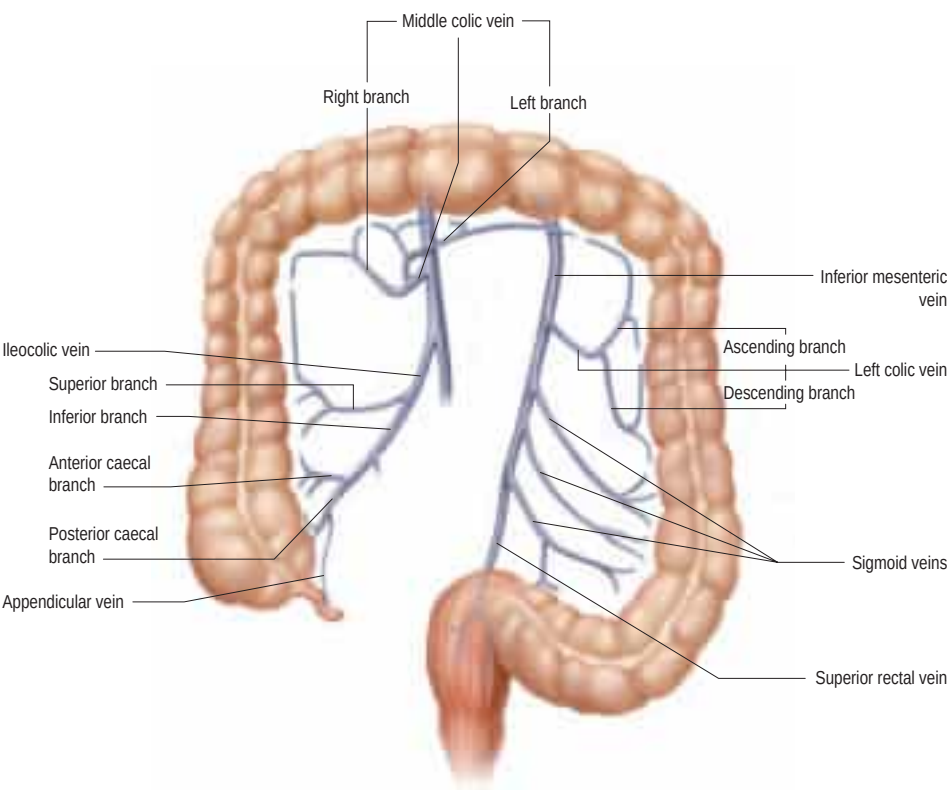


Fig. 66.15 The main branches of the superior and inferior mesenteric veins.

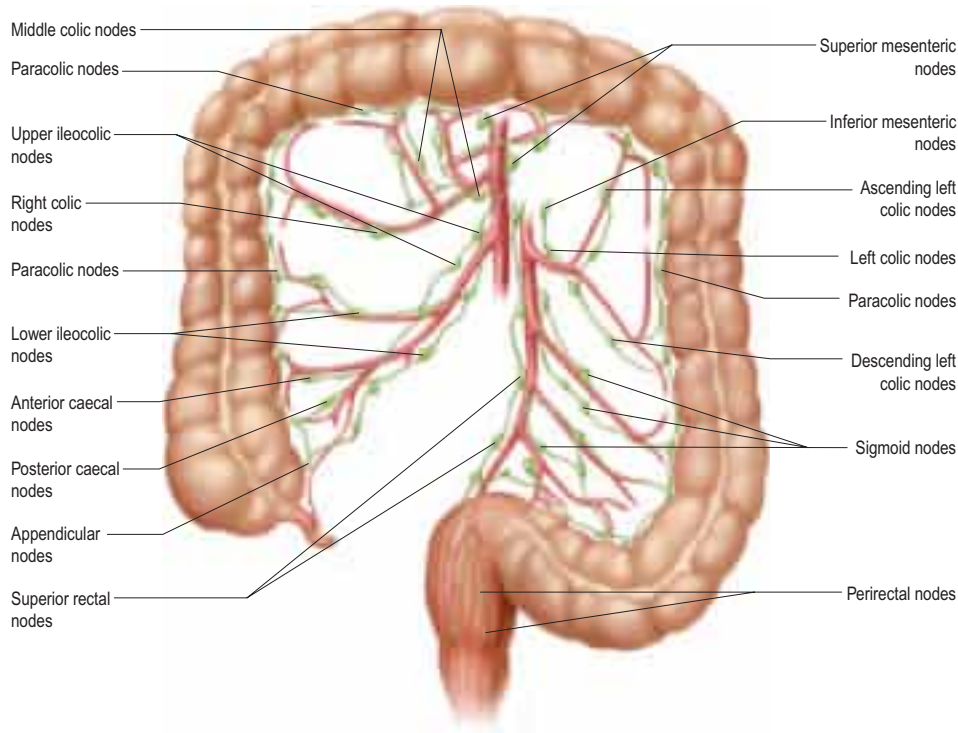


Fig. 66.16 The lymph vessels and nodes draining the large intestine.

Colic nodes

Lymph nodes related to the colon form four groups: epicolic, paracolic, intermediate colic and preterminal colic nodes. Epicolic nodes are minute whitish nodules on the serosal surface of the colon, sometimes within the appendices epiploicae. Paracolic nodes lie along the medial borders of the ascending and descending colon and along the mesenteric borders of the transverse and sigmoid colon. Intermediate colic nodes lie along the named colic vessels (the ileocolic, right colic, middle colic, left colic, sigmoid and superior rectal arteries). Preterminal colic nodes lie along the main trunks of the superior and inferior mesenteric arteries and drain into pre-aortic nodes at the origin of these vessels.

Ascending mesocolic lymph nodes are significantly larger than sigmoid mesocolic lymph nodes; this is relevant to lymph node retrieval and examination after resection for colonic cancer (Ahmadi et al 2015).

Radical lymphadenectomy during resection of colorectal cancer requires removal of the highest possible lymph nodes draining the region of colon in which the tumour is located. In cases of cancer of the rectum or sigmoid colon, this usually involves excision of the preterminal colic nodes along the inferior mesenteric artery with ligation of the artery at its root or just below the origin of the left colic artery.

MIDGUT REGION OF LARGE INTESTINE

CAECUM

This blind pouch, measuring approximately 6 cm in length, lies below the level of the ileocolic junction, anterior to the fascia covering the right iliacus and psoas major, with the lateral cutaneous nerve of the thigh interposed. The caecum usually lies adjacent to the anterior abdominal wall unless the greater omentum or loops of small bowel are interposed. The caecum is often completely covered by peritoneum, which is reflected posteriorly and inferiorly to the floor of the right iliac fossa. Peritoneal folds from the posterior caecal wall may create a variety of peritoneal recesses around the caecum that have the potential to become sites of internal herniation; a superior and inferior ileocaecal recess, a retrocaecal recess and a paracolic recess have been described (Rivkind et al 1986). The retrocaecal recess frequently contains the vermiform appendix. The three taenia coli lie anteriorly, posteromedially and posterolaterally, and converge on the base of the appendix. During childhood, the lateral caecal wall outgrows the medial wall such that the orifice of the appendix, which is originally at the apex of the caecum (Fig. 66.17), usually comes to lie slightly posteromedially.

Fluid and electrolyte reabsorption by the large intestine begins in the caecum but occurs mostly in the ascending and transverse colon. The distensible 'sac-like' morphology of the caecum allows storage of large volumes of semi-liquid chyme from the small intestine. The rela-

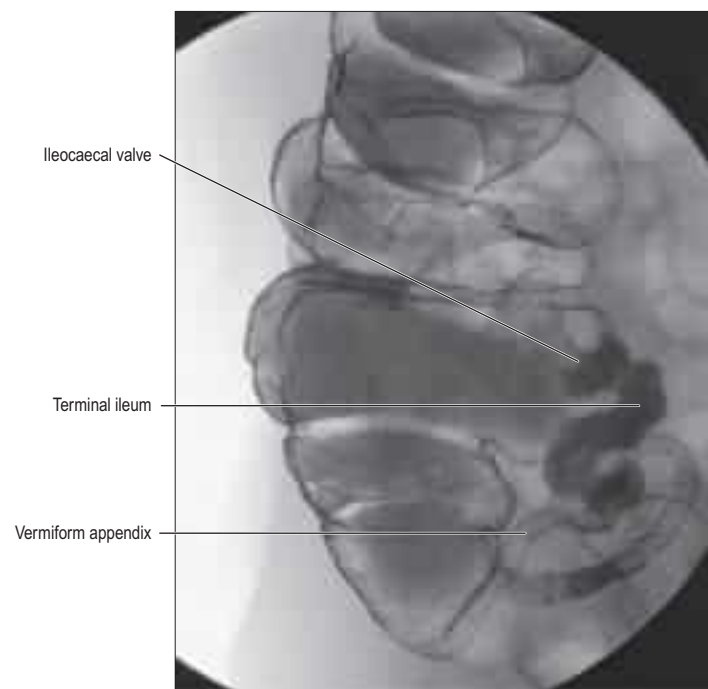


Fig. 66.17 The caecum and ileocolic junction, double contrast barium enema appearance.

tively large diameter of the caecum makes it liable to distension with increased intracolonic pressure and it is the region of the large intestine at greatest risk of perforation secondary to colonic distension (from obstruction or other pathology).

Ileocolic junction

The terminal ileum joins the posteromedial aspect of the large intestine at the junction of the caecum and colon, where it projects into the lumen of the large intestine as the ileal papilla (Fig. 66.18A). It consists of two labial folds; its precise shape and form varies from slit-like to an oval mucosal rosette, depending, in part, on the state of contraction or distension of the caecum. The upper labial fold is approximately horizontal and is at the junction between the ileum and colon; the lower lip is longer and more concave, and is at the junction between the ileum and caecum. At their bases, the labia fuse and continue as narrow mucosal ridges or frenula.

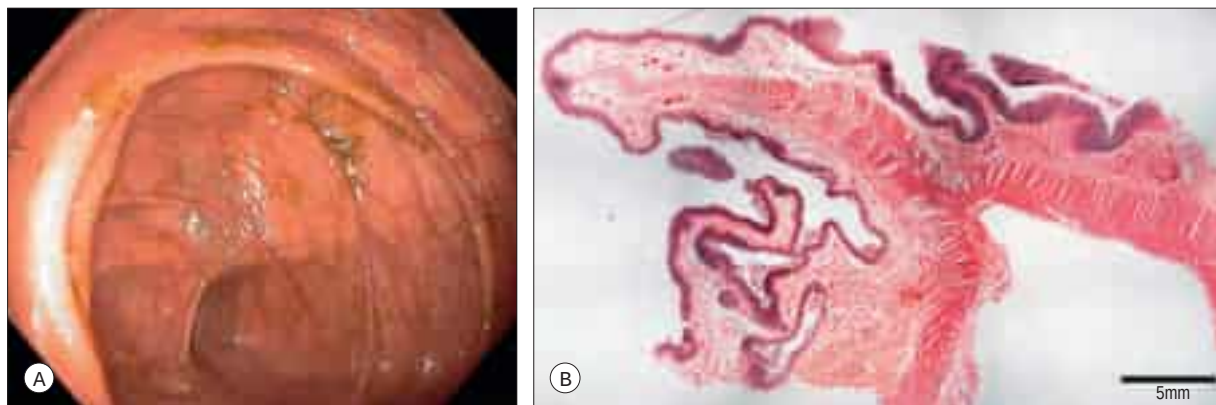


Fig. 66.18 **A**, The endoscopic appearance of the ileocolic junction ('ileocaecal valve'), which appears as a bilabial structure on the left of this view of the caecal pole. **B**, A photomicrograph showing a longitudinal section through the inferior labium of the ileal papilla (haematoxylin and eosin); the ileal mucosa is visible superiorly. (A, Courtesy of Dr Michael Schultz. B, Courtesy of Drs. Matthew Pollard and Mark Stringer.)

The ileal papilla is formed by the mucosa, submucosa and external muscle layers of the ileum, continuing through the wall of the colon and combining with corresponding layers from the caecum inferiorly and the colon superiorly (Fig. 66.18B). The internal surface of the papilla is lined by small intestinal mucosa and its colonic surface is covered by large bowel mucosa; these epithelial surfaces meet near the tip of the papilla. A localized thickening of the muscle at the base of the ileal papilla is consistent with physiological data that suggest the presence of an intrinsic anatomical sphincter (Pollard et al 2012). In addition, the bilabial configuration of the papilla may confer a valvular function. The ileocolic junction performs several roles: it provides partial mechanical and functional separation of the luminal environments of the small and large intestine, which differ in their composition, pH and bacterial content; it impedes reflux from the colon; and it helps to regulate antegrade small bowel transit.

Caecal volvulus

If the caecum and ascending colon are attached to the posterior abdominal wall by a narrow mesentery, the ileocolic region is at risk of twisting about its mesenteric pedicle, creating a caecal volvulus. In such cases, the caecum becomes markedly distended as a consequence of the strangulating closed loop bowel obstruction that develops. Non-anatomical factors such as caecal distension may also contribute to the pathogenesis of caecal volvulus (Madiba and Thomson 2002).

APPENDIX

The vermiform (worm-like) appendix is a narrow, blind-ending tube, usually between 6 and 10 cm long in the adult. It joins the postero-medial wall of the caecum below the ileocolic junction. The appendix grows in length and diameter during early childhood, reaching almost mature dimensions by about 3 years of age (Searle et al 2013). The appendix usually lies in the right iliac fossa but its tip may occupy one of several positions (Fig. 66.19) (Buschard and Kjaeldgaard 1973). In clinical practice, the tip is most commonly retrocaecal or retrocolic (behind the caecum or lower ascending colon, respectively, anterior to iliacus and psoas major), or pelvic (when the appendix descends over the pelvic brim, in close relation to the right uterine tube and ovary in females). Other positions include subcaecal, and pre- or post-ileal (anterior or posterior to the terminal ileum, respectively), especially when a long appendicular mesentery allows greater mobility. The surface marking for the base of the appendix has traditionally been described by McBurney's point (two-thirds of the way along a line between the umbilicus and right anterior superior iliac spine) but its position is variable and affected by posture, caecal distension and other factors (Hale et al 2010).

The appendix has a continuous outer layer of longitudinal muscle formed by the coalescence of the three taeniae coli. Its lumen is irregularly narrowed by submucosal lymphoid tissue. The mesoappendix is a triangular mesentery running between the terminal ileum and appendix; it contains a variable amount of fat and frequently ends short of the tip of the appendix. A small fold of peritoneum runs between the terminal ileum and the anterior layer of the mesoappendix (the so-called 'bloodless fold of Treves'), and another fold of peritoneum containing the anterior caecal vessels extends from the terminal ileal mesentery to the anterior wall of the caecum (and contains the anterior caecal

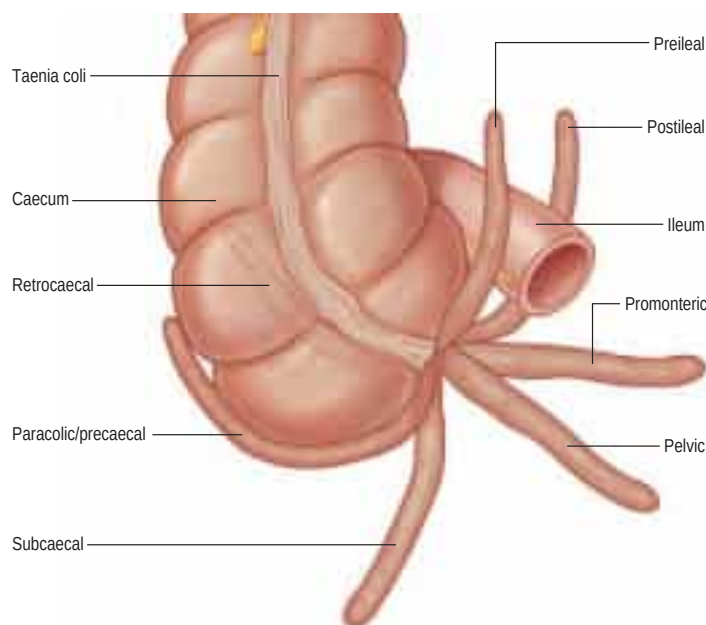


Fig. 66.19 The major positions of the appendix encountered at surgery or postmortem.

artery) (see Fig. 63.10). The luminal orifice is sometimes partially covered by a mucosal fold forming an asymmetrical 'valve' (see Fig. 66.5). The lumen may be widely patent in early childhood but is often partially or wholly obliterated in the elderly. Agenesis or duplication of the appendix are exceptionally rare (Barlow et al 2013).

Microstructure

The layers of the wall of the appendix are similar to those of the large intestine in general but with some notable differences. The serosa forms a complete covering, except along the mesenteric attachment. The outer longitudinal muscle is a complete layer of uniform thickness, except in a few small areas where the muscularis externa is deficient, allowing the serosa to come into contact with the submucosa.

The submucosa typically contains large lymphoid aggregates that may extend into the mucosa and disrupt the integrity of the muscularis mucosae (Fig. 66.20). The mucosa is covered by a columnar epithelium, which contains M cells where it overlies the mucosal lymphoid tissue. Glands (crypts) (see Fig. 4.5) are similar to those of the colon but are fewer in number and less densely packed. They penetrate deep into the lamina propria. The submucosal lymphoid tissue frequently exhibits germinal centres within its follicles (see Fig. 4.10), indicative of B-cell activation. Lymphoid follicles are absent at birth but accumulate during the first 10 years of life to become prominent. In adults, the lymphoid follicles gradually atrophy; in the elderly, the lumen of the appendix may be partially obliterated by fibrous tissue.

Acute appendicitis



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Acute appendicitis may develop as a consequence of obstruction of the lumen from inspissated material, a faecolith (appendicolith) or lymphoid swelling. This can lead to suppurative, infarction and necrosis. Appendicoliths are more common in children than adults but their prevalence is variable, reflecting differences between populations, definitions and methods of detection. In one radiographic study of normal appendices removed incidentally at surgery or at autopsy, calcified faecoliths were identified in 2.7% (Felson 1949). The increased size of the appendicular orifice in early childhood and the decreased lumen in the elderly may be reasons why acute appendicitis is less common in these age groups. Although the appendix is well supplied by arterial anastomoses at its base, the appendicular artery is an end artery; its close proximity to the wall of the appendix makes it susceptible to thrombosis during acute appendicitis, which explains the high frequency of gangrenous perforation seen in the disease.

Visceral afferent nerves are responsible for the initial symptoms of acute appendicitis arising from distension and inflammation of the organ: namely, colicky pain with or without vomiting. These afferent nerves enter the spinal cord at around the level of the tenth thoracic spinal segment. Abdominal pain from appendicitis is poorly localized initially and referred to the central (periumbilical) region of the abdomen, consistent with the midgut origin of the appendix. It is not until parietal tissues adjacent to the appendix become involved in the inflammatory process that somatic nociceptors are stimulated, resulting in localization of pain to the right iliac fossa. Removal of the appendix is not by itself associated with any discernible long-term sequelae but this does not mean that it is an entirely vestigial organ. There is some evidence that the appendix acts as a reservoir for normal gut flora, enabling the large bowel flora to recover more rapidly after severe gastroenteritis (Randal Bollinger et al 2007).

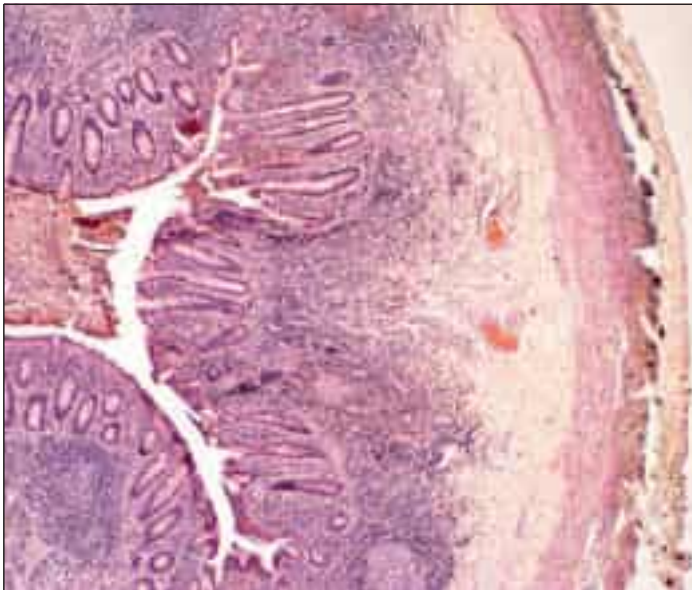


Fig. 66.20 A low-power micrograph of the appendix in transverse section, showing part of its circumference and a faecal pellet lodged in its lumen. Lymphoid tissue (basophilic staining) occupies much of the mucosa between crypts, and part of the submucosa. The muscularis externa and outermost serosal layer are seen at the right of the field.

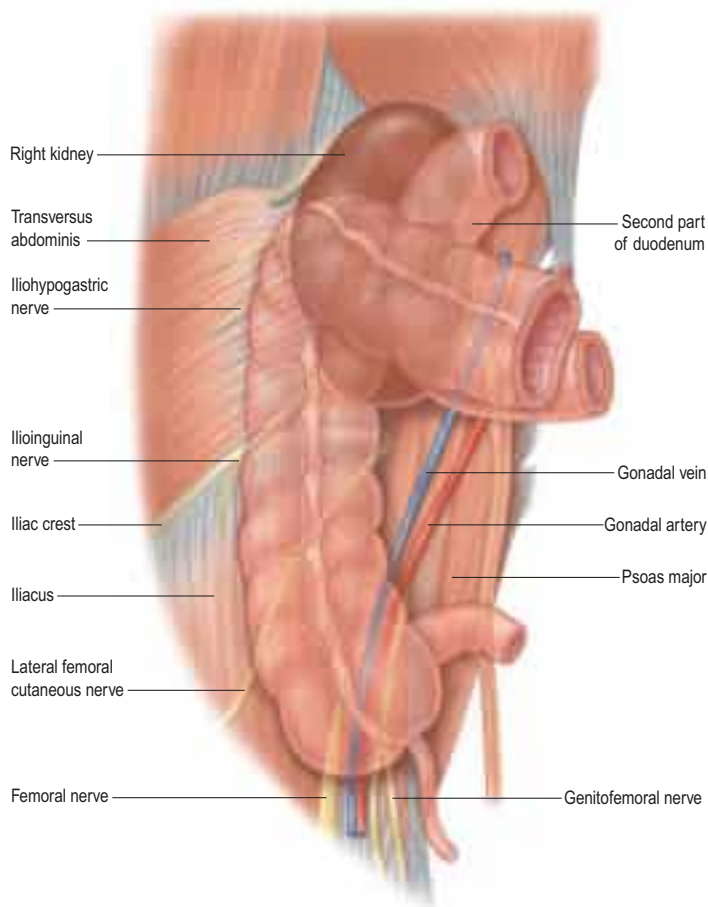


Fig. 66.21 Posterior relations of the ascending colon.

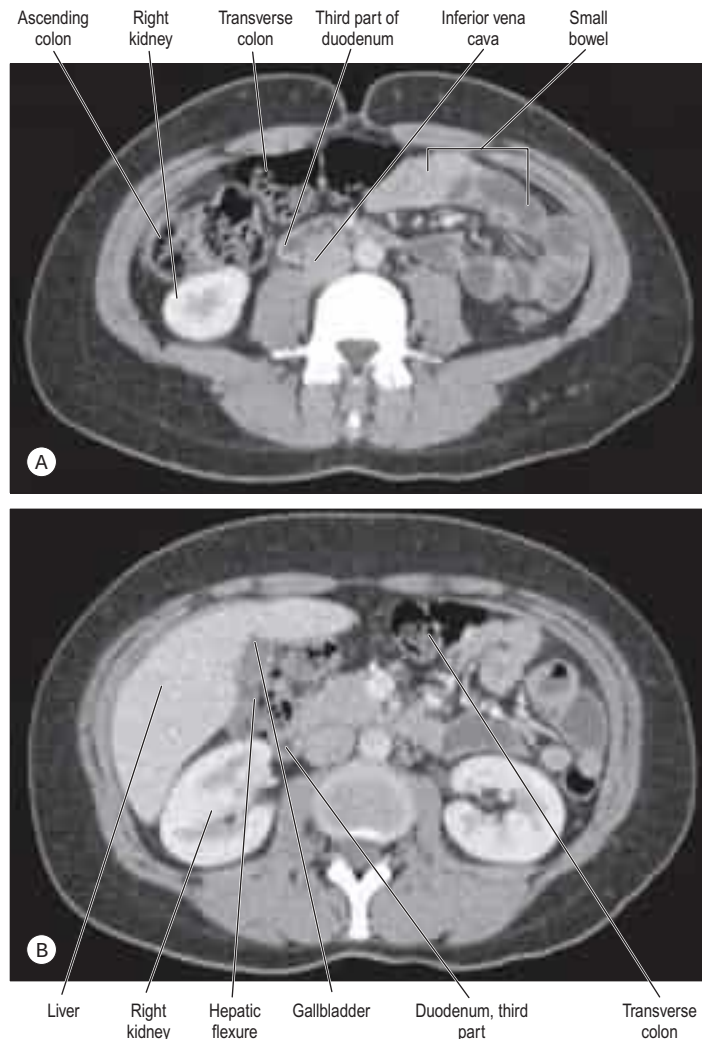


Fig. 66.22 Axial CT images of the ascending colon obtained (A) at the level of the mid ascending colon and (B) at the level of the hepatic flexure, showing the relationship of the ascending colon and hepatic flexure to the right lobe of the liver, the right kidney, the second part of the duodenum, and the gallbladder. (Courtesy of Dr Louise Moore, Chelsea and Westminster Hospital, London.)

of the ascending colon are covered by peritoneum, which runs laterally into the right paracolic gutter and medially into the right infracolic compartment. The ascending mesocolon is composed of two layers of mesothelium containing fat embedded within a connective tissue lattice, vessels and nerves; it is attached to the retroperitoneum of the posterior abdominal wall via a layer of connective tissue (Coffey 2013, Culligan et al 2014), known as Toldt's fascia, which forms the plane of dissection when performing a right hemicolectomy.

Hepatic flexure

The hepatic flexure, forming the junction between the ascending and transverse colon, is variable in position, and has a less acute angle than the splenic flexure. It overlies the anterior surface of the lower pole of the right kidney, abutting the inferior surface of the right lobe of the liver above (see Fig. 66.21; Fig. 66.22B), the second part of the duodenum medially, and the fundus of the gallbladder anteromedially. The posterior aspect of the hepatic flexure is not covered by peritoneum and is in direct contact with renal fascia. The greater omentum often extends from its attachment to the transverse colon on to the hepatic flexure.

TRANSVERSE COLON

The transverse colon is intraperitoneal. It is highly variable both in length (approximately 50 cm long on average) and the extent to which it hangs down anterior to the small bowel between sites of attachment at the right (hepatic) and left (splenic) colic flexures. The greater curvature of the stomach and the gastrocolic omentum, which fuses with the

ASCENDING COLON

The ascending colon is 15–20 cm long and passes upwards from the ileocolic junction to the right colic (hepatic) flexure, separated posteriorly by loose connective tissue from the iliac fascia, iliolumbar ligament, quadratus lumborum, transversus abdominis, and the renal fascia anterior and inferolateral to the right kidney. The lateral femoral cutaneous nerve, usually the fourth lumbar artery, and sometimes the ilioinguinal and iliohypogastric nerves lie posteriorly as they cross quadratus lumborum (Fig. 66.21). Both the lateral and anterior surfaces

Occasionally, the ascending mesocolon is reflected forwards from the posterior abdominal wall, conferring greater mobility on this usually retroperitoneal part of the large bowel ([Saunders et al 1995](#)).

transverse colon anteriorly and continues inferiorly as the greater omentum, are superior. The transverse colon is suspended by the transverse mesocolon, which is attached from the lower pole of the right kidney, across the second part of the duodenum and pancreas, to the upper pole of the left kidney (see Fig. 63.2). The splenic flexure lies at a higher level than the hepatic flexure, often abutting the spleen under the left lower ribs. The disposition of the transverse colon and more posteriorly sited flexures results in the anterior taenia of the ascending (and descending) colon lying inferiorly (see above). The transverse mesocolon affords considerable mobility to the transverse colon. Occasionally, the colon may be interposed between the liver and the diaphragm, when it may be mistaken for free intraperitoneal gas on imaging; when associated with gastrointestinal symptoms it is referred to as Chilaiditi's syndrome (Murphy et al 2000).

VASCULAR SUPPLY AND LYMPHATIC DRAINAGE OF THE MIDGUT

Arteries

Superior mesenteric artery

The artery of the midgut is the superior mesenteric artery, which arises from the anterior surface of the aorta at the level of the lower border of the body of the first lumbar vertebra. It runs steeply downwards, posterior to the splenic vein and body of the pancreas, with the superior mesenteric vein on its right, and directly anterior to the left renal vein, the uncinate process of the pancreas and the third part of the duodenum. It then enters the root of the mesentery of the small intestine and passes obliquely downwards and to the right, giving off several branches to the large intestine.

Ileocolic artery

The ileocolic artery arises from the superior mesenteric artery near the root of the mesentery of the small intestine, descending within the mesentery to the right towards the caecum, and crossing anterior to the right ureter, gonadal vessels and psoas major. It usually divides into superior and inferior branches, the superior branch running up along the left side of the ascending colon to anastomose with the right colic artery (or right branch of the middle colic artery) (Veeresh et al 2012). The inferior branch runs to the ileocolic junction and divides into anterior and posterior caecal arteries, the appendicular artery, and an ileal branch that passes to the left in the ileal mesentery to anastomose with a terminal ileal branch of the superior mesenteric artery (Fig. 66.23). The latter therefore provides a collateral arterial supply to the caecum. The ileocolic artery provides the major arterial supply to the caecum; traction on the caecum in the direction of the anterior superior iliac spine will cause the artery to tent up the mesentery, allowing easy identification of the vessel.

The appendicular artery usually arises directly from the ileocolic artery and descends posterior to the terminal ileum to enter the meso-appendix a short distance from the base of the appendix (see Fig. 66.23). Here, it gives off a recurrent branch, which anastomoses at the base of the appendix with a branch of the posterior caecal artery. The appendicular artery approaches the tip of the organ, at first near to, and then in the edge of, the mesoappendix. The terminal part of the artery lies on the wall of the appendix and may become thrombosed in appendicitis, resulting in distal gangrene or necrosis. Less commonly, the appendicular artery may arise from the posterior caecal artery or an

ileal artery (Ouattara et al 2007) (see Fig. 66.12). Accessory appendicular arteries are common; two or more arteries may supply the appendix.

Right colic artery

The right colic artery is relatively small and variable in its anatomy (Batra et al 2013). It usually arises as a common trunk with the middle colic artery, but may originate directly from the superior mesenteric artery, or from the ileocolic artery (when it is referred to as an accessory right colic artery) (Fig. 66.24). It passes to the right, across the right psoas major and quadratus lumborum, crossing the right gonadal vessels and ureter, just posterior to the peritoneal floor of the right infracolic compartment. Near the left side of the ascending colon, it divides into a descending branch, which runs down to anastomose with the superior branch of the ileocolic artery, and an ascending branch, which passes up across the lower pole of the right kidney to the hepatic flexure, where it anastomoses with a branch of the middle colic artery. Together, these anastomoses form the marginal artery at the hepatic flexure.

Middle colic artery

The middle colic artery arises from the right side of the superior mesenteric artery, either separately or in common with the right colic artery, just inferior to the neck of the pancreas, and passes anteriorly and superiorly within the transverse mesocolon, just to the right of the midline. As it approaches the colon, it usually divides into right and left branches. The right branch anastomoses with the ascending branch of the right colic artery. The left branch supplies the terminal part of the midgut and anastomoses with a branch of the left colic artery near the splenic flexure. The marginal artery thus formed lies a few centimetres from the mesenteric edge of the transverse colon. Sometimes, the middle colic artery divides into three or more branches within the

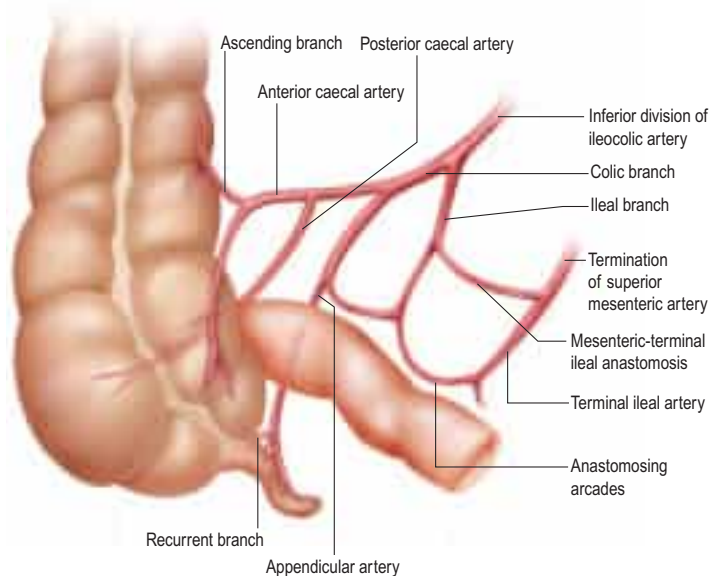


Fig. 66.23 The arteries of the caecum, vermiform appendix and ascending colon.

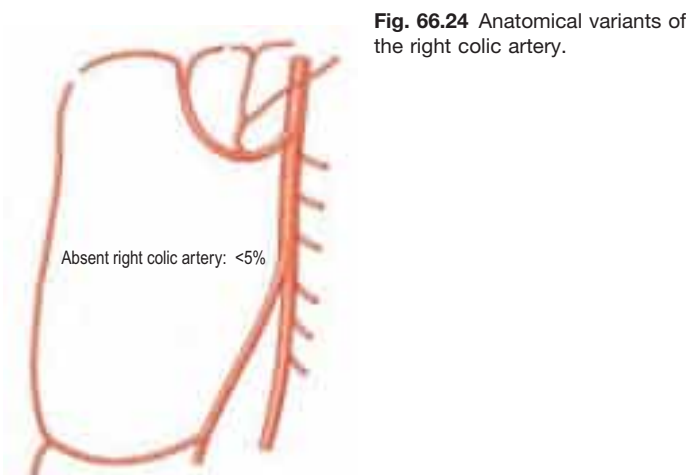
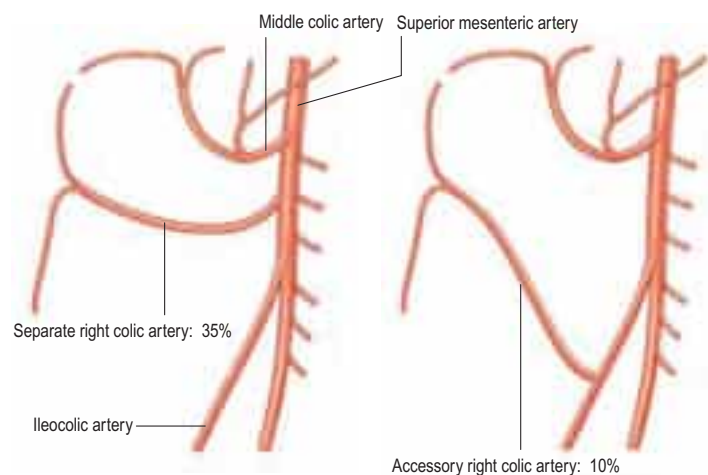


Fig. 66.24 Anatomical variants of the right colic artery.

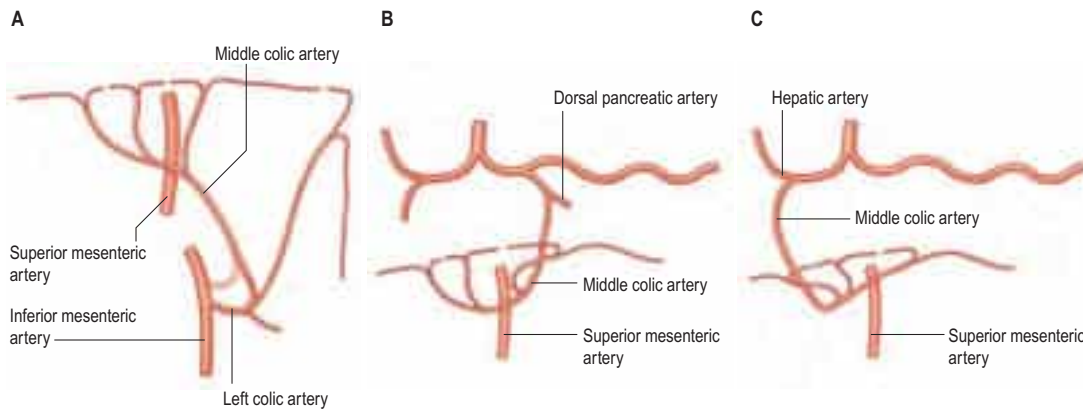


Fig. 66.25 Variants in the origin of the middle colic artery; only replaced arteries have been illustrated for simplicity. Accessory arteries are more common than completely replaced arteries. **A**, The left colic artery or inferior mesenteric artery. **B**, The dorsal pancreatic artery. **C**, The hepatic artery. Each variant accounts for fewer than 5% of cases.

transverse mesocolon, in which case the most lateral branches form the arterial anastomoses. An accessory, or rarely a replaced, middle colic artery may arise directly from the aorta (Yoshida *et al* 1993), dorsal pancreatic, hepatic, inferior mesenteric or left colic arteries (Fig. 66.25). In addition, an accessory middle colic artery is occasionally found arising from the superior mesenteric artery proximal to the origin of the actual middle colic artery (Turmezei and Cockburn 2009).

Veins

All the branches of the superior mesenteric artery are accompanied by correspondingly named veins. These tributaries drain into the superior mesenteric vein, which ascends to the right of the artery, crossing the third part of the duodenum and uncinate process of the pancreas. Behind the neck of the pancreas, at the level of the transpyloric plane (the lower border of the body of the first lumbar vertebra), it joins the splenic vein to form the portal vein, which ascends behind the first part of the duodenum to reach the liver. The right colic vein is highly variable: it may drain either directly into the superior mesenteric vein, or into the right gastroepiploic or inferior pancreaticoduodenal veins to form a 'gastrocolic trunk' that drains into the superior mesenteric vein, or it may be entirely absent (Yamaguchi *et al* 2002). Several tributaries draining into one or more middle colic veins are highly variable in extent and position. The middle colic veins drain either into the superior mesenteric vein, just before its junction with the splenic vein, or directly into the hepatic portal vein. The appendicular vein usually joins the caecal vein to become the ileocolic vein; infection from the appendix can therefore be carried directly to the liver via the portal vein.

Lymphatic drainage

Lymphatic vessels originate from both anterior and posterior aspects of the colon. Lymph drainage from the midgut follows the course of the superior mesenteric artery. Thus, lymph from the caecum and appendix drains to nodes associated with the ileocolic artery, and from the distal ascending colon and hepatic flexure to nodes along the right colic artery (see Fig. 66.16). When the distal transverse colon and splenic flexure are predominantly supplied by the middle colic artery, lymph drainage of this segment is also largely to superior mesenteric nodes. Lymphatic vessels in the appendix are numerous and combine to form vessels that ascend in the mesoappendix, occasionally interrupted by one or more nodes. They unite to form three or four larger vessels that anastomose with lymphatic vessels draining the ascending colon into ileocolic nodes.

HINDGUT REGION OF LARGE INTESTINE

Splenic flexure

The splenic flexure marks the junction between the transverse and descending colon, and lies in the left hypochondrium, anterior to the tail of the pancreas and the left kidney (Fig. 66.26). Its position with respect to the spleen is variable: it usually lies inferomedial to the lower pole, forming the colic impression, but it may lie anterior to the splenic hilum, or even a little above. The splenic flexure is often attached to the splenic capsule by a peritoneal ligament and inadvertent downward traction on the flexure during surgery may tear the splenic capsule. The phrenicocolic ligament attaches the flexure to the diaphragm below the inferior pole of the spleen at about the level of the tenth rib. The splenic flexure often adopts a very acute angle such that the end of the transverse colon overlaps the beginning of the descending colon; there may

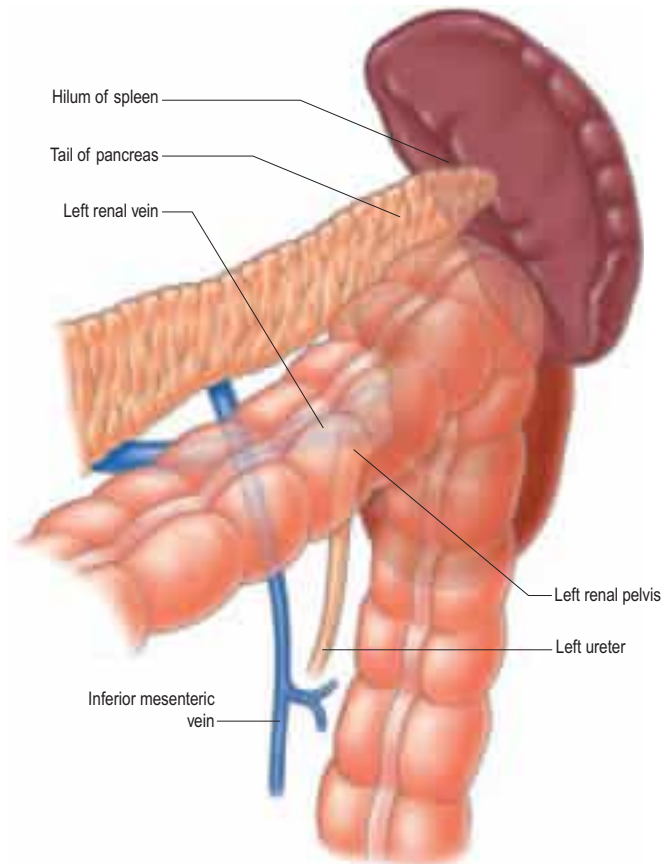


Fig. 66.26 Relations of the splenic flexure.

be peritoneal and omental adhesions between the two structures. It lies more superiorly and posteriorly than the hepatic flexure.

DESCENDING COLON

The descending colon is 25–30 cm long and descends from the splenic flexure in the left hypochondrium to the level of the iliac crest, where it curves medially anterior to iliacus to become the sigmoid colon. In most adults it is retroperitoneal, covered anteriorly and on both sides by peritoneum, but occasionally the descending colon is more mobile, being suspended from the posterior abdominal wall by a short mesocolon (Saunders *et al* 1995). Its lateral peritoneal reflection in the left paracolic gutter is marked by a white line (of Toldt). The posterior surface of the descending colon is separated by a layer of loose connective tissue from the anterior renal fascia inferolateral to the left kidney, transversus abdominis, quadratus lumborum, iliacus and the lateral margin of psoas major (Fig. 66.27). The subcostal vessels and nerves, iliohypogastric, ilioinguinal, lateral femoral cutaneous, femoral and genitofemoral nerves, and, usually, the fourth lumbar artery also lie posteriorly, and loops of jejunum are anterior. The descending colon is smaller in calibre and more deeply placed than the ascending colon. Appendices epiploicae are more common in this part of the colon.

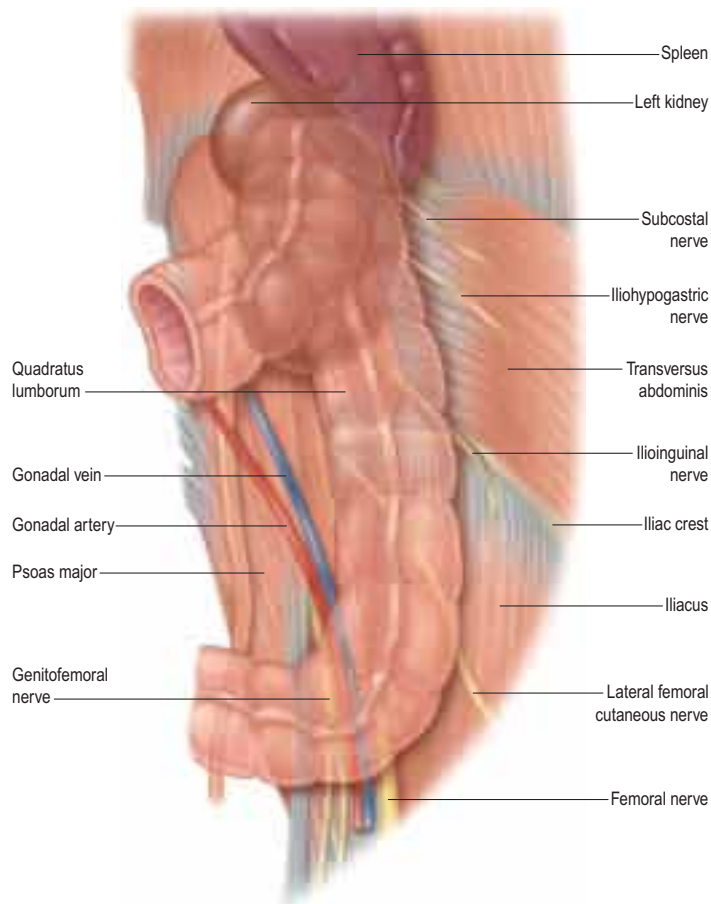


Fig. 66.27 Posterior relations of the descending colon.

SIGMOID COLON

The sigmoid colon runs from the lesser pelvis to the beginning of the rectum at the level of the third sacral vertebra. It is usually suspended from the posterior abdominal and pelvic walls by a fan-shaped sigmoid mesocolon, but may be tethered by congenital adhesions to the parietal peritoneum over iliacus. Consequently, its length and position are highly variable. It usually lies within the pelvic cavity anterior to the rectum on the peritoneal surface of the bladder (and uterus in the female). It is usually completely invested in peritoneum. The root of the sigmoid mesocolon has an inverted V-shaped attachment; the right limb of this inverted 'V' ascends from a point anterior to the third sacral vertebra to the bifurcation of the left common iliac vessels (crossed by the left ureter) and descends from this point along the external iliac vessels.

Between the fixed junctions with the descending colon and rectum, the relations of the sigmoid colon are quite variable (Fig. 66.28). Laterally are the left external iliac vessels, the obturator nerve, ovary or vas deferens, and the lateral pelvic wall; posteriorly lie the left external and internal iliac and gonadal vessels, ureter, piriformis and the sacral plexus; anteroinferiorly lie the bladder in males, or the uterus and bladder in females; superiorly and to the right, the sigmoid colon is in contact with loops of the ileum. The gonadal vessels and ureter lie in a distinct fascial plane, which is the inferior extension of the retroperitoneal perirenal fascia and is separate from the sigmoid mesocolon. This plane can be recognized and separated during surgical resection of the sigmoid mesocolon. The taeniae coli of the sigmoid colon are wider than elsewhere in the colon, and coalesce at its distal end to form a complete circumferential longitudinal muscle layer. Appendices epiploicae are particularly prominent in the sigmoid colon.

The position and shape of the sigmoid colon depend on its length (which increases with age) and mobility; the length of the sigmoid mesocolon (usually longer in males); and the degree of distension of the colon, rectum, bladder and uterus. The length of the sigmoid colon and its mesentery vary between ethnic groups (Alatise et al 2013).

Sigmoid volvulus



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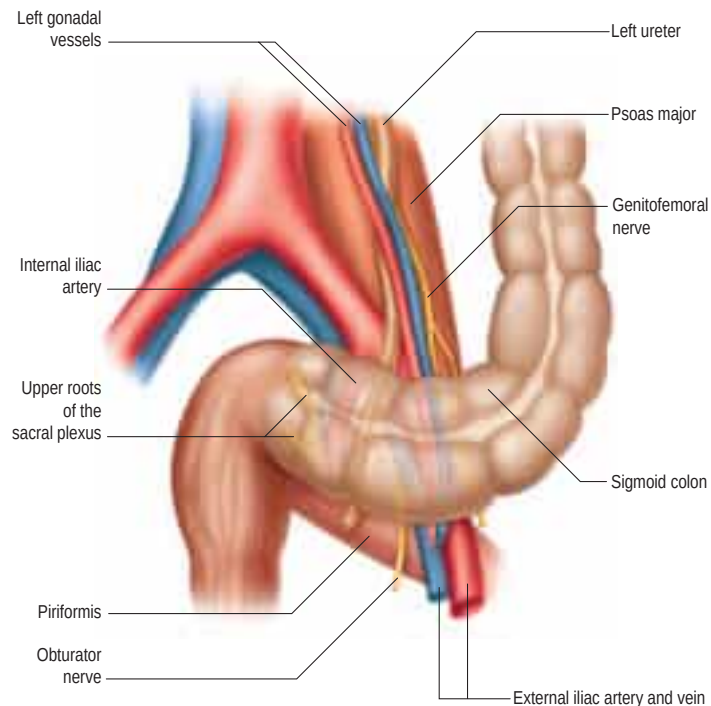


Fig. 66.28 Posterior relations of the sigmoid colon.

Diverticular disease



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MESOCOLON

The mesocolon extends along the entire length of the colon and is continuous with the small bowel mesentery proximally and the mesorectum distally. The mesocolon consists of fat embedded within a connective tissue lattice and sandwiched between two layers of mesothelium; it also contains nerves, blood and lymphatic vessels, and lymph nodes. A layer of loose connective tissue, known as Toldt's fascia, lies immediately posterior to the mesocolon, where it is adherent to the retroperitoneum of the posterior abdominal wall (Culligan et al 2014). Complete excision of the relevant segment of mesocolon by dissecting within or behind the plane of Toldt's fascia has been shown to improve survival in colon cancer (Søndenaa et al 2014).

RECTUM

The rectum is continuous with the sigmoid colon at the level of the third sacral vertebra and ends as it passes through the pelvic floor (the puborectalis part of levator ani), where it is continuous with the anal canal. It descends within the sacrococcygeal concavity, at first running posteriorly and then curving anteriorly. The junction between the anteriorly directed distal rectum and the posteriorly directed anal canal is the anorectal angle, which is maintained by puborectalis. This is 2–3 cm anterior to and slightly below the tip of the coccyx, level with the apex of the prostate in males. The rectum also deviates in three lateral curves: upper, convex to the right; middle (the most prominent), convex to the left; and lower, convex to the right. Both ends of the rectum are in the median plane (Fig. 66.30).

Although variable in absolute length, the rectum is often defined in clinical practice as extending approximately 15 cm above the external anal margin. Its upper diameter is similar to that of the sigmoid colon, but more inferiorly it becomes dilated as the rectal ampulla. Unlike the sigmoid colon, the rectum has no sacculations, appendices epiploicae or taeniae coli. The taeniae merge a few centimetres above the recto-sigmoid junction, initially forming wide anterior and posterior muscular bands that then fuse to form a continuous outer layer of longitudinal muscle investing the entire length of the rectum. At the rectal ampulla, a few longitudinal muscle fibres pass forwards from the anterior rectal wall to the perineal body and urethra (rectourethralis).

The upper third of the rectum is covered by peritoneum on its anterior and lateral aspects, and the middle third by peritoneum on its anterior aspect only; the lower third is below the peritoneum. The

Rotation or volvulus of the sigmoid colon may occur around its mesenteric attachment. Volvulus is more likely when the length of the sigmoid colon and the length and width of its mesentery are greater (Akinkuotu et al 2011). Volvulus does not occur when the sigmoid colon and its mesocolon are short. The anatomical features predisposing to sigmoid volvulus are most commonly found in sub-Saharan Africans and chronically institutionalized patients.

Acquired diverticula commonly occur in the sigmoid colon, particularly in Western populations. These diverticula frequently develop in parallel rows between the mesenteric taenia coli and the two anti-mesenteric taeniae where the colonic wall is weaker; at this site, the outer longitudinal muscle is deficient and the circular muscle is traversed by arteries supplying the submucosal vascular plexus (Slack 1962). However, the predilection of diverticula in the sigmoid colon probably relates to causative factors rather than intrinsic differences in the structure of this segment of the colon. In contrast, congenital diverticula may occur at any site in the colon (Fig. 66.29) and are often found along the mesenteric border.



Fig. 66.29 A CT scan (coronal reformat) showing both right colonic and sigmoid diverticulosis. (Courtesy of Dr Kamini Patel, Homerton Hospital, London.)

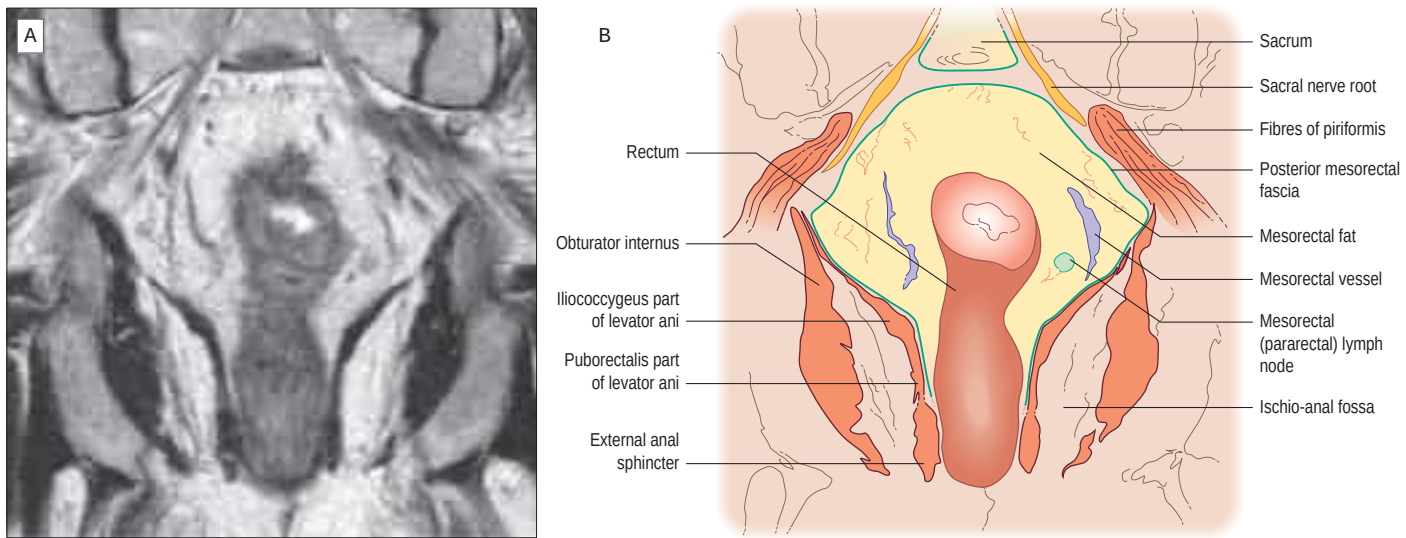


Fig. 66.30 **A**, A coronal T2-weighted MRI of the rectum. **B**, A line diagram illustrating the main features in **A**.

peritoneum is reflected anteriorly on to the urinary bladder in males to form the rectovesical pouch, or on to the posterior vaginal fornix in females to form the recto-uterine pouch (of Douglas). The level of this reflection is higher in males; the rectovesical pouch is approximately 7.5 cm above the anorectal junction in males, while the recto-uterine pouch is approximately 5.5 cm above the anorectal junction in females. In neonatal boys, the peritoneum extends on the anterior rectal wall as far as the lower limit of the prostate. Superiorly, the peritoneum is firmly attached to the muscle layer of the sigmoid colon by fibrous connective tissue, but as it descends on to the rectum, it is more loosely attached by fatty connective tissue, which allows considerable expansion of the upper half of the rectum.

When the rectum is empty, the mucosa has several longitudinal folds in its lower part, which become effaced during distension. The rectum commonly has three permanent semilunar transverse or horizontal folds (although the number can vary), which are most marked in rectal distension (see Fig. 66.10). Two types of horizontal fold have been recognized. One consists of the mucosa, the circular muscle layer and part of the longitudinal muscle, and is marked externally by an indentation. The other is devoid of longitudinal muscle and has no external marking. The most superior fold at the beginning of the rectum may be on either the left or the right and occasionally encircles the rectal lumen. The middle fold is largest and most constant, and lies immediately above the rectal ampulla, projecting from the anterior and right wall just below the level of the anterior peritoneal reflection; the circular muscle is particularly prominent in this fold. The inferior fold is on the left and the most variable.

Relations Posterior to the rectum and mesorectum, separated by the presacral fascia, are the lower three sacral vertebrae, coccyx, median and lateral sacral vessels, and the lowest portion of the sacral sympathetic chain. Laterally, the upper rectum is related to the sigmoid colon and/or small bowel, while below the peritoneal reflection lie levator ani, obturator internus, the obturator nerve and vessels, ureters, the inferior hypogastric plexus, internal iliac vessels, piriformis and the sacral plexus. Anteriorly, the rectum is related to the sigmoid colon and/or small bowel in both sexes, and the base of the bladder, seminal vesicles, vas deferens, terminal parts of the ureters, and the prostate in males (see Figs 66.34, 76.17). In females, the cervix/body of the uterus and vagina are anterior, the latter separated from the rectum by a rectovaginal septum. In postmenopausal females, and after childbirth, the connective tissue of the rectovaginal septum may atrophy or become thinned, reducing the support of the anterior rectal and posterior vaginal walls.

MESORECTUM AND RECTAL FASCIAE

Although the rectum has no mesentery, its surrounding fat is enclosed within a fascial envelope known as the mesorectum. This constitutes a distinct compartment that is intimately related to the rectum down to the level of levator ani (Figs 66.31–66.36). It contains the superior rectal artery and its branches, the superior rectal vein and its tributaries, lymphatic vessels and nodes, and adipose connective tissue. The mesorectum is much bulkier posteriorly (see Figs 66.31, 66.32). It is enclosed

by mesorectal fascia, a distinct layer of connective tissue that is also called the fascia propria of the rectum. This fascia limits the mesorectum posteriorly and lies anterior to the presacral fascia (of Waldeyer), separated by a so-called retrorectal 'space'. The rectum and mesorectal fascia are connected to the parietal fascia of the pelvis by two 'lateral ligaments' (see below) and one posterior ligament, the rectosacral fascia (Gaudio et al 2010). The latter is a thin layer of connective tissue that passes obliquely downwards from the presacral fascia to fuse with the mesorectal fascia posterior to the rectum 3–5 cm above the anorectal junction (García-Armengol et al 2008). The mesorectal fascia is surrounded by a thin and relatively avascular layer of loose areolar tissue, which separates it from the posterior and lateral walls of the true pelvis. Superiorly, the mesorectal fascia blends with the connective tissue bounding the sigmoid mesentery. Laterally, it extends around the rectum and mesorectum, and becomes contiguous anteriorly with a denser condensation of fascia, conveniently described eponymously as Denonvilliers' fascia. This fascia is sometimes confusingly known as rectovesical fascia in males but it extends caudal to the bladder and is particularly dense posterior to the prostate and seminal vesicles; it lies immediately anterior to the mesorectal fascia, to which it may be fused (Lindsey et al 2000). In females, Denonvilliers' fascia combines with the anterior mesorectal fascia to form the rectovaginal septum (Zhai et al 2009).

Mesorectal fascia is visible by CT, MRI and dissection (Bissett et al 2001). On MRI scans, the mesorectum appears as a fat-filled envelope containing blood vessels and lymph nodes; small nerves are not visible but interlacing connective tissue strands can be seen. The mesorectal fascia is recognizable on axial MRI images as a low-signal layer surrounding the mesorectum, corresponding to a distinct condensation of fascia seen on histological sections (Brown et al 1999). Identification of this layer in patients with a malignant rectal tumour can help predict the success of surgical resection. Branches of the inferior hypogastric plexus and middle rectal vessels enter the mesorectum anterolaterally. They are ensheathed by fascia and are sometimes collectively referred to as the 'lateral rectal ligaments'. The number and calibre of the middle rectal vessels are highly variable; they may be very small or even absent (Sato and Sato 1991). The 'lateral ligaments' are not clearly seen on MRI or CT scanning, and only appear as an identifiable structure with surgical traction on the rectum. The fascia of the 'ligaments' is flimsy and probably plays little role in support of the rectum. Parietal fascia covers levator ani and the muscles of the side wall of the pelvis, and is continuous posteriorly with the presacral fascia (see Fig. 66.36).

Rectal prolapse



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Rectocele



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Rectal cancer



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In prolapse of the rectum involving all layers of the rectal wall, the mid and upper rectum descend towards the pelvic floor on straining. Descent occurs within the lumen of the rectum as a form of intussusception, and may be a consequence of the large diameter of the rectal ampulla and relative fixation of the anal canal. The loose adipose tissue of the upper mesorectum and the 'lateral ligaments' offer only limited resistance against rectal descent. Chronic enlargement of the anorectal space bound by levator ani commonly occurs in patients suffering from recurrent rectal prolapse, but is probably a consequence rather than a cause of the prolapse. In females, the recto-uterine pouch also descends with the anterior rectal wall.

When the rectovaginal septum is grossly effaced, especially in postmenopausal females, the pressure of defecation can cause bulging of the rectal wall into the posterior vagina and, in extreme cases, through the vaginal introitus. Weakness of the pelvic floor muscles contributes to the prolapse by allowing descent of the perineum during straining.

An important concept in the surgical treatment of adenocarcinoma of the rectum is the necessity of excising the rectum in continuity with the mesorectum and its contained lymph nodes in order to avoid local

tumour recurrence ([Heald et al 1982](#)). In total mesorectal excision, the posterolateral plane of dissection is outside the mesorectal fascia but anterior to the presacral fascia and its underlying venous plexus. Laterally, the plane is interrupted only by small branches of the middle rectal vessels; anteriorly, it merges with the rectovesical or rectovaginal fascia. The inferior hypogastric plexus is closely related to the plane of dissection laterally; these nerves must be preserved to retain bladder and sexual function. Total mesorectal excision involves complete excision of the rectum down to the level of the pelvic floor, where the puborectalis component of levator ani merges with the deep component of the external anal sphincter. Superiorly, the inferior mesenteric artery is ligated near its origin.

In abdominoperineal excision of the rectum, it should be noted that the mesorectum becomes extremely thin at the point where the rectum passes through levator ani. To achieve adequate surgical margins, the plane of dissection is through the ischio-anal fossa and levator ani outside the puborectalis sling and into the mesorectal plane well above the pelvic floor.

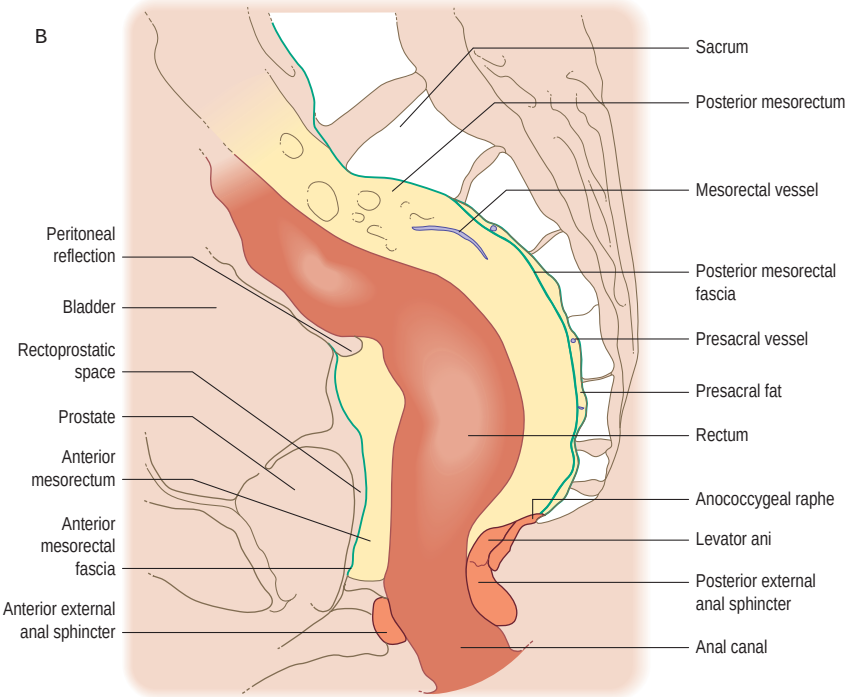
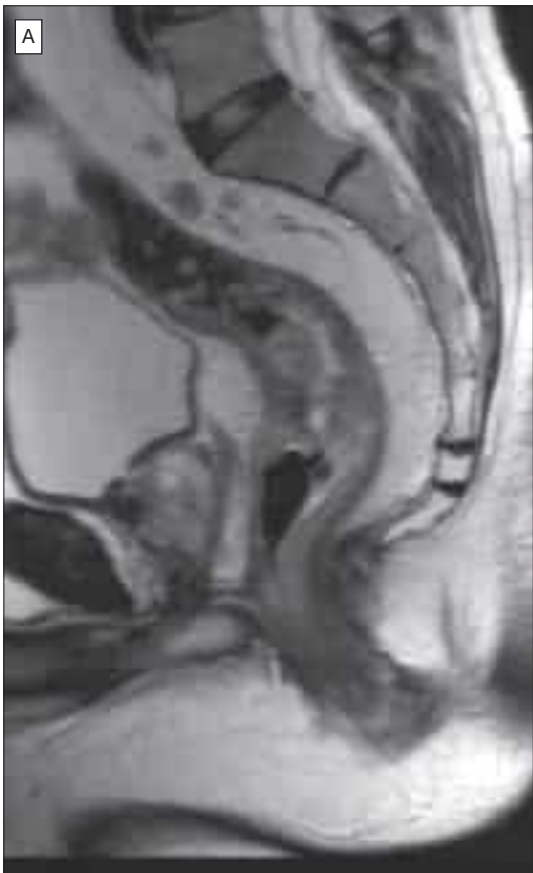


Fig. 66.31 A, A sagittal T2-weighted MRI of the rectum in a male. B, A line diagram illustrating the main features in A.

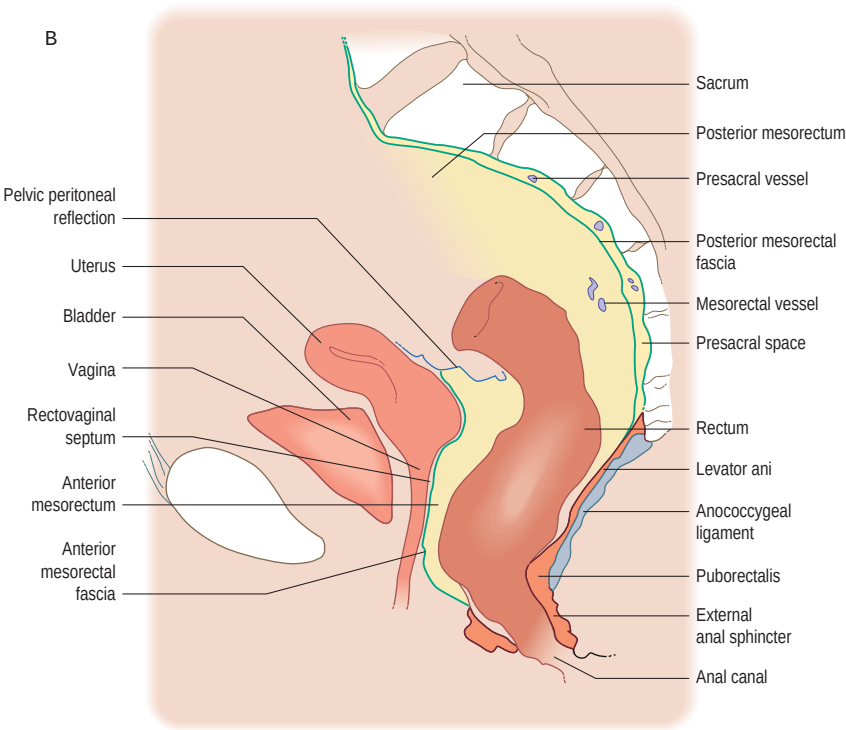
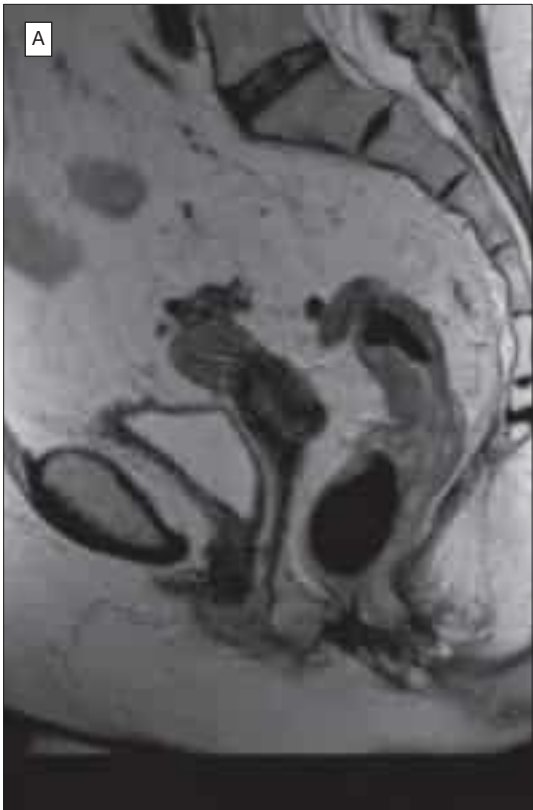


Fig. 66.32 A, A sagittal T2-weighted MRI of the rectum in a female. B, A line diagram illustrating the main features in A.

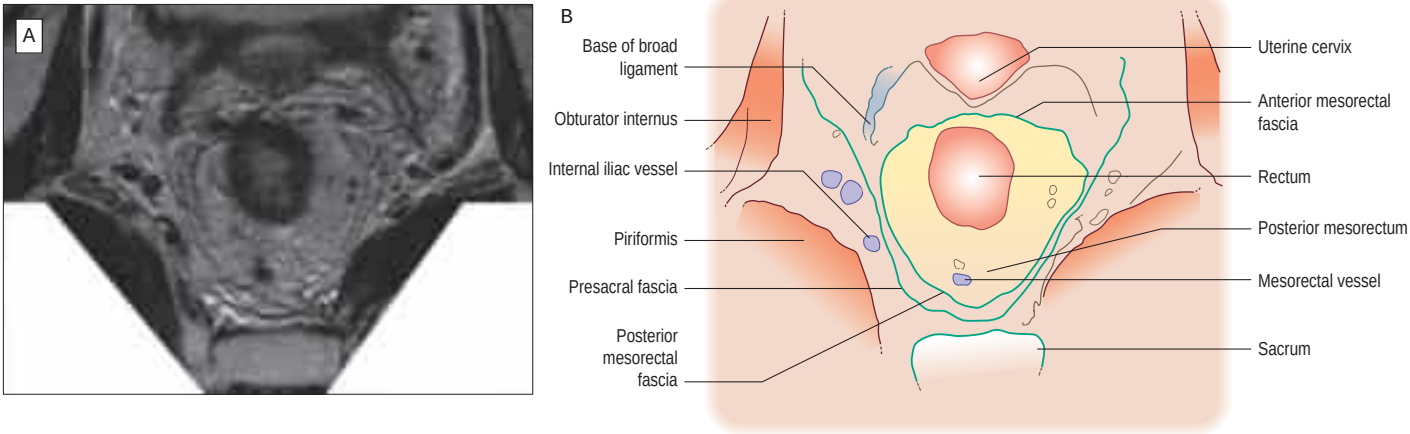


Fig. 66.33 **A**, An axial T2-weighted MRI of the upper rectum in a female. **B**, A line diagram illustrating the main features in **A**.

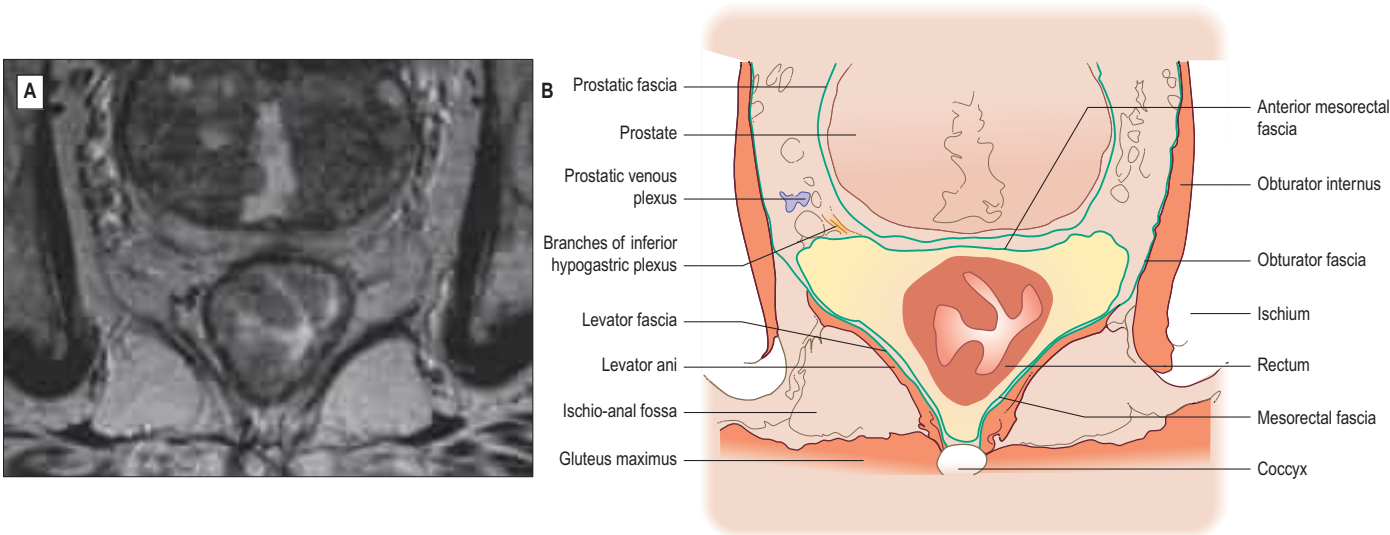


Fig. 66.34 **A**, An axial T2-weighted MRI of the mid rectum below the peritoneal reflection in a male. **B**, A line diagram illustrating the main features in **A**.

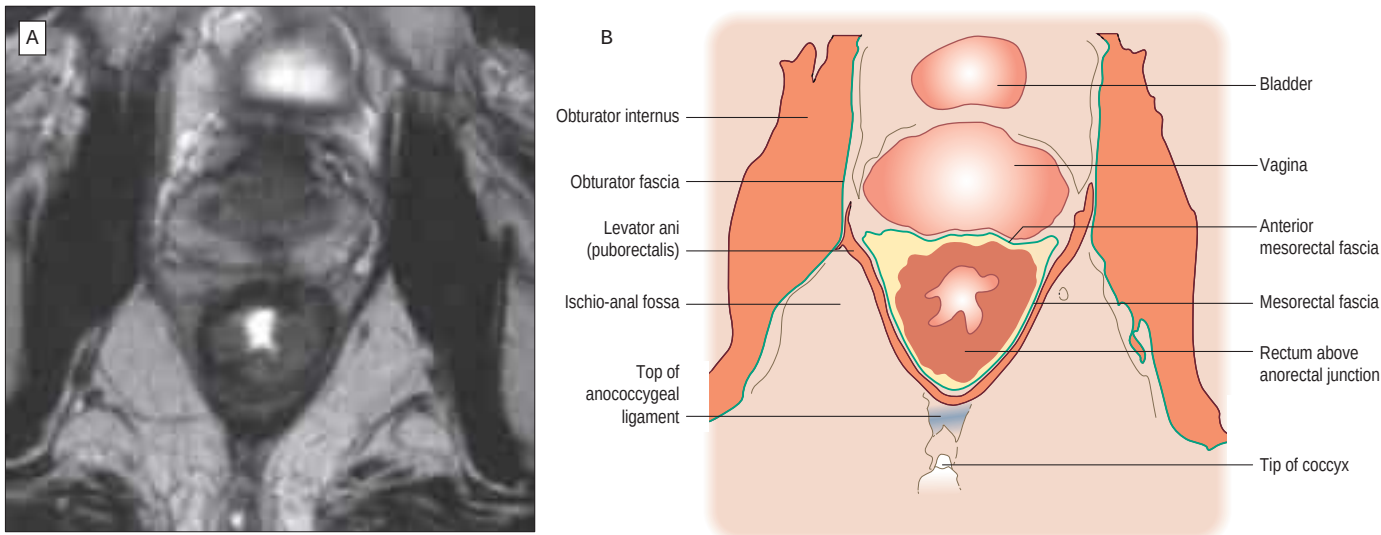


Fig. 66.35 **A**, An axial T2-weighted MRI of the low rectum below the peritoneal reflection in a female. **B**, A line diagram illustrating the main features in **A**.

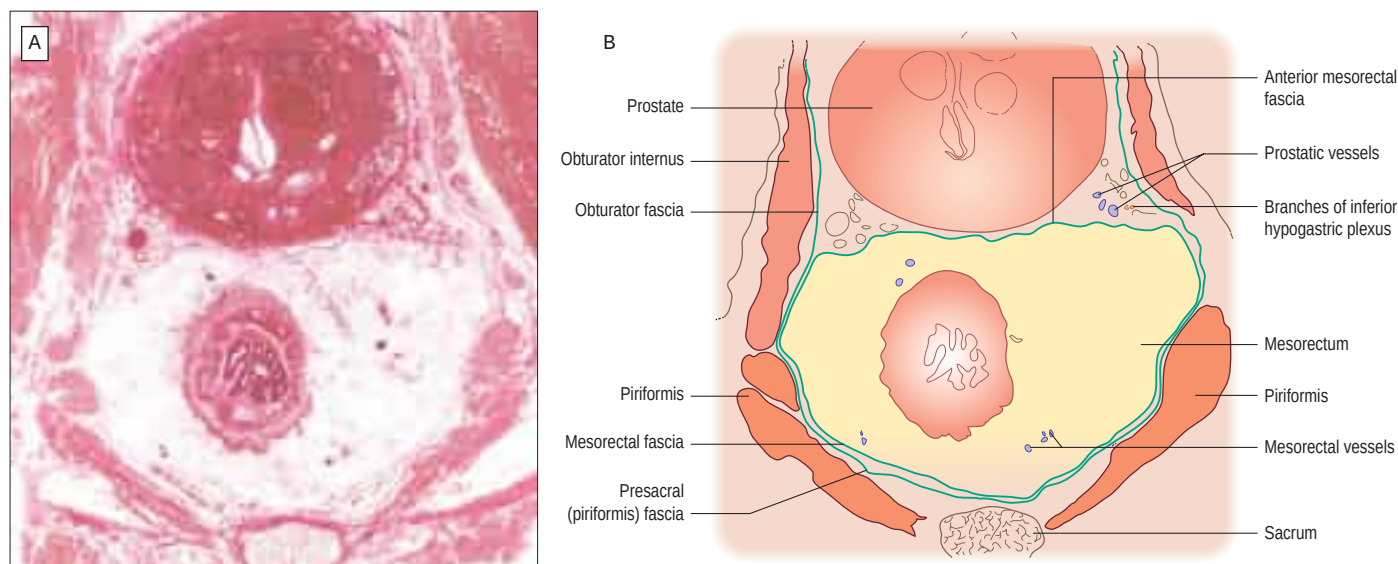


Fig. 66.36 **A**, A light microscope transverse section of the mid rectum (male, cadaveric specimen). **B**, A line diagram illustrating the main features in **A**.

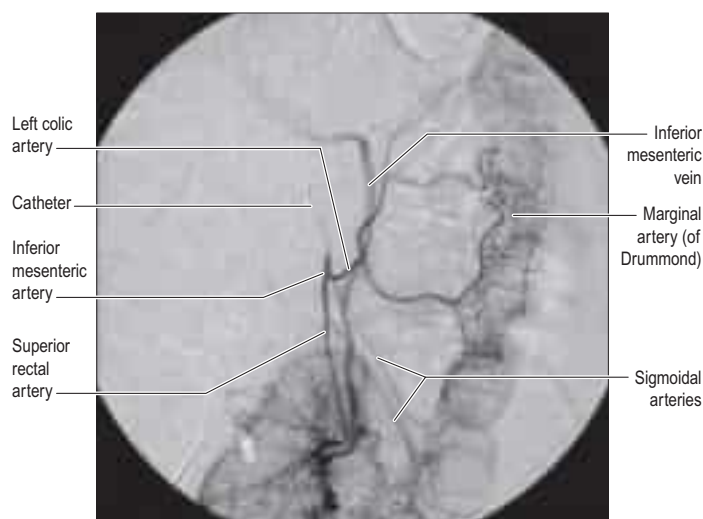


Fig. 66.37 A digital subtraction arteriogram showing the inferior mesenteric artery and its branches.

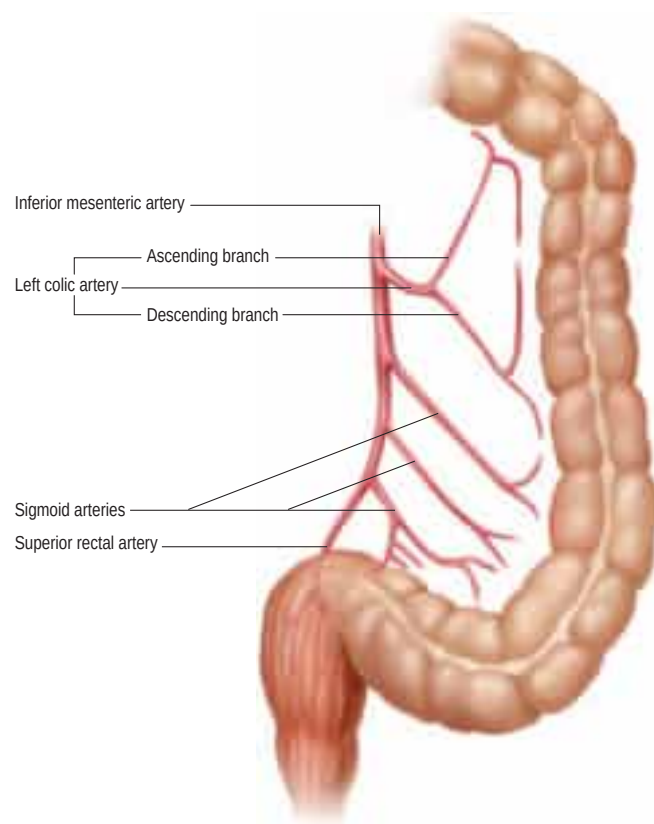


Fig. 66.38 The inferior mesenteric artery.

VASCULAR SUPPLY AND LYMPHATIC DRAINAGE OF THE HINDGUT

Arteries

Inferior mesenteric artery

The artery of the hindgut is the inferior mesenteric artery (**Figs 66.37–66.39**), which arises from the anterior or left anterolateral aspect of the aorta behind the inferior border of the third part of the duodenum 3–4 cm above the aortic bifurcation, at the level of the third lumbar vertebra. It runs obliquely down to the pelvic brim, beneath the peritoneal floor of the left infracolic compartment, initially anterior and then to the left of the aorta. It gives off the left colic and sigmoid arteries, and crosses the origin of the left common iliac artery medial to the ureter, with the inferior mesenteric vein lying between. Beyond the pelvic brim, it continues in the root of the sigmoid mesocolon as the superior rectal artery.

Left colic artery

The left colic artery usually arises from the inferior mesenteric artery shortly after its origin, ascends within the left colic mesentery and divides into an ascending and a descending branch (see **Figs 66.38–66.39**). The ascending branch passes upwards across the left psoas major, gonadal vessels, ureter and left kidney, and is crossed by the inferior mesenteric vein; its terminal branches anastomose with those of the left branch of the middle colic artery within the transverse mesocolon. The descending branch passes laterally and downwards, and anastomoses with branches from the ascending branch and the highest sigmoid artery to form part of the marginal artery. The arterial arcades

thus formed supply the distal third of the transverse and the descending colon. The left colic artery may originate from or in common with a sigmoid artery (**Murono et al 2015**). Occasionally, an accessory, or rarely a replaced, left colic artery may originate from the trunk of the superior mesenteric artery or its middle colic or first jejunal branch (**Fig. 66.40**). When present, it runs laterally in the upper left colic mesentery just inferior to the duodenojejunal flexure to supply the upper descending colon, and forms part or all of the marginal artery in the region of the distal transverse colon. The left colic artery may itself give rise to an accessory left middle colic artery. Occasionally, the left colic artery gives rise to a branch shortly after its origin, which ascends in the mesentery and anastomoses directly with a similar descending branch of the left branch of the middle colic artery (the so-called arc of Riolo; **van Gulik and Schoots 2005**).

The dominant arterial supply of the splenic flexure is usually from the left colic artery but may be from the left branch of the middle colic artery. The marginal artery in this region may be absent or small, but it may enlarge considerably if the inferior mesenteric artery is stenosed or occluded (see **p. 1100**).

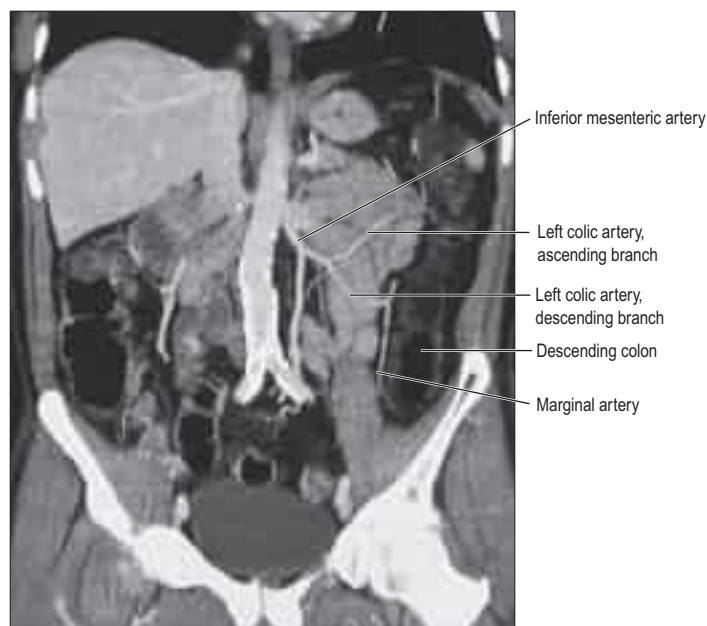


Fig. 66.39 The vascular supply of the descending colon from the inferior mesenteric artery via the ascending and descending branches of the left colic artery, coronal reformat CT. (Courtesy of Dr Louise Moore, Chelsea and Westminster Hospital, London.)

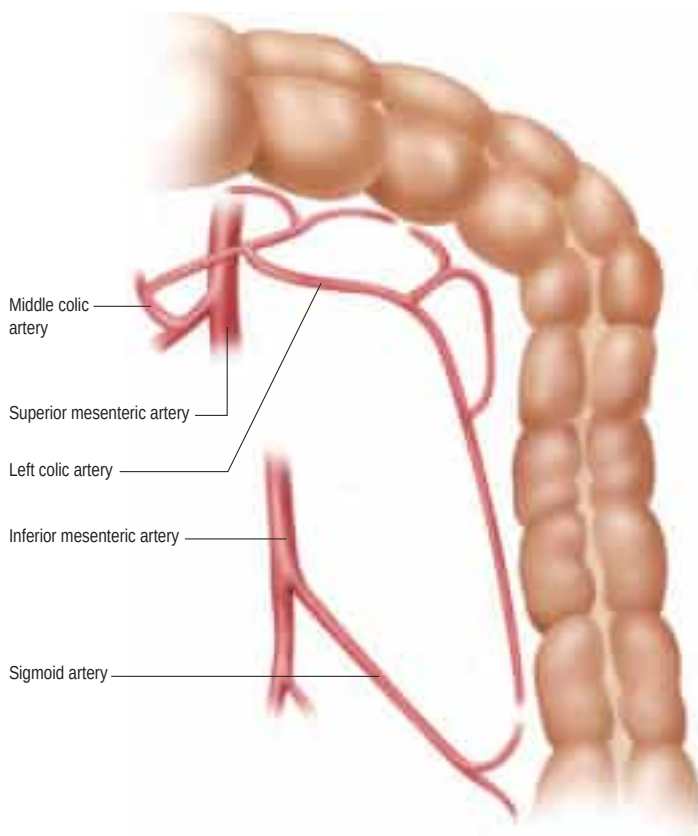


Fig. 66.40 A replaced left colic artery arising from the middle colic branch of the superior mesenteric artery.

Sigmoid arteries

The inferior mesenteric artery gives rise to between two and five sigmoid arteries, which descend obliquely in the sigmoid mesocolon anterior to the left psoas major, ureter and gonadal vessels. They supply the distal descending colon and sigmoid colon, and anastomose superiorly with the left colic artery and inferiorly with the superior rectal artery. Unlike the arrangement in the small intestine, arterial arcades do not form until the arteries are close to the wall of the colon, when small branches arise that supply the sigmoid colon directly. A true marginal artery is less obvious in the sigmoid colon. A significant interval often exists in the mesentery between the highest sigmoid artery and the descending branch of the left colic artery; this forms a useful guide to the arterial territories during surgical dissection.

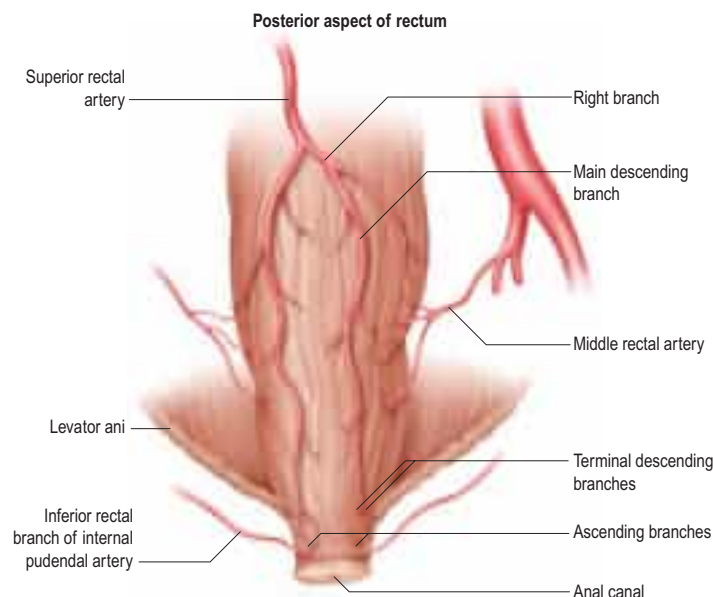


Fig. 66.41 The arterial supply of the rectum.

Vascular ligation in left colonic resections

During resection of the distal descending and sigmoid colon, ligation of the inferior mesenteric artery close to its origin preserves the bifurcation of the left colic artery and helps to maintain the arterial supply to the proximal descending colon via the anastomosis between the left branch of the middle colic artery and the ascending branch of the left colic artery. Ligation of the left colic artery close to its bifurcation may interfere with this supply and render the proximal descending colon more likely to become ischaemic. Similarly, if the inferior mesenteric vein is ligated, then the bifurcation of the left colic vein forms the route of venous drainage from the proximal descending colon to the middle colic vein.

Superior rectal artery

The principal arterial supply to the upper two-thirds of the rectum is via the artery of the hindgut – specifically, its pelvic continuation, the superior rectal artery (**Fig. 66.41**). The inferior mesenteric artery crosses the left common iliac vessels medial to the ureter and descends in the medial limb of the sigmoid mesocolon, straddled by the inferior hypogastric nerves on either side. As it crosses the pelvic brim, it becomes the superior rectal artery. At the level of the third sacral vertebra, where the rectum begins, the artery enters the upper mesorectum in the midline and divides into two branches that descend, initially posterolaterally, and then on each side of the rectum. Terminal branches pierce the rectal wall and anastomose with branches of the middle and inferior rectal arteries within the rectal submucosa.

Middle and inferior rectal arteries

The middle rectal arteries arise either directly from the anterior division of the internal iliac artery or from the inferior vesical artery (vaginal artery in females). When present, they enter the mesorectum anterolaterally in the 'lateral ligaments' and provide some additional supply to the middle third of the rectum. The inferior rectal arteries are terminal branches of the internal pudendal arteries. They cross the ischio-anal fossa to enter the upper anal canal laterally and supply the internal and external anal sphincters, the anal canal and perianal skin. Ascending branches supply the distal third of the rectum, anastomosing with terminal branches of the superior rectal artery in the rectal submucosa. The rectum also receives a small arterial supply from the median sacral artery via a branch that enters posteriorly at the level of the anorectal junction.

Veins

Inferior mesenteric vein

The inferior mesenteric vein drains the rectum, sigmoid, descending and distal transverse colon (see **Fig. 66.15**; **Fig. 66.42**). It begins as the continuation of the superior rectal vein from the rectal plexus, through which it connects with the middle and inferior rectal veins. The inferior mesenteric vein lies to the left of the inferior mesenteric artery in the retroperitoneum and ascends anterior to the left psoas major and left ureter; it may cross the gonadal vessels or ascend medial to them. It lies just lateral or occasionally posterior to the duodenojejunal flexure, where it can be located intraoperatively. The inferior mesenteric vein



Fig. 66.42 The inferior mesenteric vein and its tributaries, digital subtraction arteriogram. (Courtesy of Dr Adam Mitchell, Charing Cross Hospital, London.)

receives the superior rectal vein, several sigmoid veins, and the left colic vein. It usually passes posterior to the lower border of the body of the pancreas and anterior to the left renal vein to drain into the splenic vein, but it may drain into the confluence of the splenic and superior mesenteric veins or directly into the superior mesenteric vein (Graf et al 1997).

Rectal venous plexus

A rectal venous plexus surrounds the rectum and connects anteriorly with the vesical plexus in males or the uterovaginal plexus in females. It consists of internal veins beneath the mucosa of the rectum and upper anal canal, and external veins lying outside the muscular wall. The upper two-thirds of the rectum and the internal part of the rectal venous plexus drain mainly to the superior rectal vein and, from there, to the inferior mesenteric vein; the middle third of the rectum drains by one or more middle rectal veins into the internal iliac vein; and the lower third of the rectum and anal canal drains via inferior rectal veins into the internal pudendal veins. The rectal venous plexus is therefore a site of communication between the portal and systemic venous systems.

Superior rectal vein

Venous tributaries from the rectal venous plexus ascend in the rectal submucosa and pierce the rectal wall to form the superior rectal vein. This runs in the upper mesorectum and root of the sigmoid mesocolon to the left of the superior rectal artery, crossing the pelvic brim and left common iliac vessels to form the inferior mesenteric vein (see Figs 66.15, 66.44).

Left colic vein

The left colic vein is formed from several tributaries, including ascending and descending branches that correspond to similarly named arteries. The left colic vein usually lies superior to its corresponding artery, and has a shorter course because the inferior mesenteric vein lies lateral to the inferior mesenteric artery (see Fig. 66.42). Occasionally, there are two left colic veins that both drain into the inferior mesenteric vein.

Lymphatic drainage

Small lymphoid aggregates measuring no more than a few millimetres are occasionally visible on the surface of the large bowel (epicolic and epirectal nodes). Lymph from the descending colon drains to paracolic nodes in the mesentery adjacent to the bowel and, from there, to intermediate nodes along the left colic artery (Jamieson and Dobson 1909). Lymphatics from the sigmoid colon follow a similar pathway from

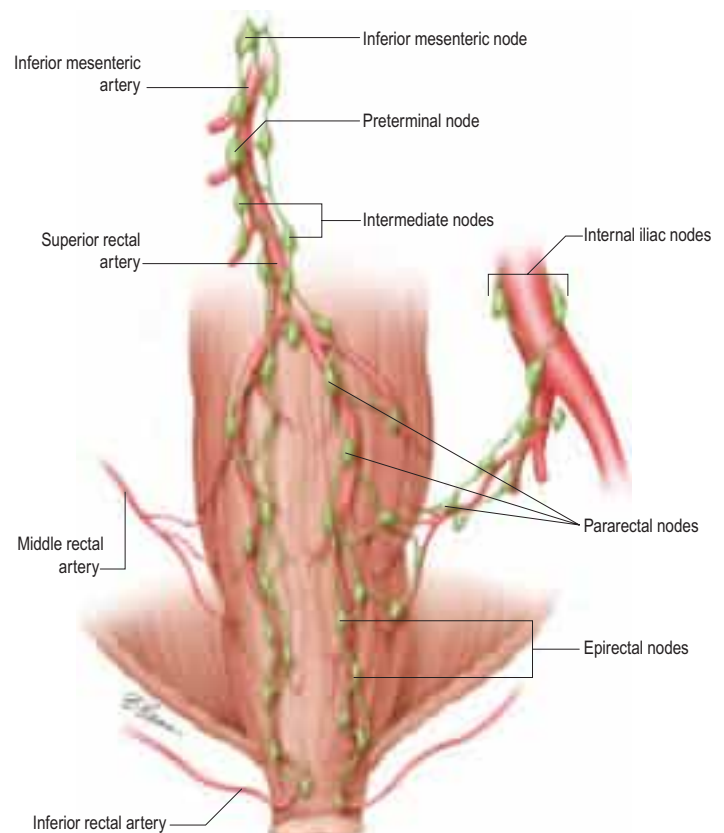


Fig. 66.43 Lymph nodes of the rectum and upper anal canal (viewed from behind).

paracolic to intermediate nodes in the sigmoid mesocolon. Intramural lymph from the rectum and anal canal above the dentate line drains predominantly to pararectal nodes in the adipose tissue of the mesorectum and then along lymphatics accompanying the superior rectal artery (Fig. 66.43). These intermediate nodes then drain to lymph nodes along the inferior mesenteric artery and, from there, to pre-aortic nodes. Some lymphatics from the lower rectum travel with the middle and inferior rectal arteries to internal iliac nodes, and along the median sacral artery to presacral nodes. The clinical significance of such drainage in relation to the spread of malignancy from the lower rectum is debated (Bell et al 2009).

INNERVATION OF THE LARGE INTESTINE

The innervation of the large intestine is complex, and includes the enteric nervous system (made up of motor neurones, intrinsic sensory neurones, and interneurones lying within the wall of the gut); the autonomic nervous system (sympathetic and parasympathetic innervation); and extrinsic sensory innervation (visceral afferents) (Brookes et al 2009) (Ch. 59).

The enteric nervous system consists of ganglionated nerve plexuses lying in the submucosa (Meissner's plexus) and between the longitudinal and circular smooth muscle layers (Auerbach's myenteric plexus). Collectively, these nerves are concerned with control of mucosal functions (secretion, blood flow) and the propulsion of luminal contents by rhythmic and synchronized contractions. Motor neurones may be inhibitory or excitatory. Intrinsic sensory neurones are activated by mechanical stimuli such as stretch and muscle tension, and by chemical stimuli released by neuroendocrine cells in the wall of the gut (see Fig. 3.26). Sensory neurones project locally on to myenteric interneurones and motor neurones, allowing the spread of reflex activity along the gut. Interstitial cells of Cajal are present within the submucosal and myenteric nerve plexuses. In summary, they act as a link between smooth muscle cells and extrinsic neurones, and are responsible for the intrinsic pacemaker activity of the gut; their activity is modulated by autonomic nerves.

The large intestine is richly innervated by sympathetic neurones that originate either directly from the sympathetic chain or indirectly via the aortic plexus; the former are mostly involved in the control of blood flow (vasoconstriction), while the latter also influence the secretory activity and motility of the gut. The cell bodies of preganglionic sympathetic fibres supplying the midgut are found in the intermediolateral

columns of the fifth to the twelfth thoracic spinal segments and those of the hindgut in the intermediolateral columns of the first and second lumbar spinal segments. Postganglionic sympathetic neurones from both sources release noradrenaline (norepinephrine), causing presynaptic inhibition within enteric circuits, slowing gut motility and driving contraction of the ileocaecal and internal anal sphincters. Sympathetic supply to the midgut is conveyed to the coeliac and superior mesenteric plexuses via the greater and lesser splanchnic nerves; postganglionic axons are distributed with branches of the superior mesenteric artery. Sympathetic supply of the hindgut is conveyed via the lumbar splanchnic nerves that synapse in the abdominal aortic and inferior mesenteric plexuses, and via sacral splanchnic nerves that synapse in the superior and inferior hypogastric plexuses; postganglionic fibres are distributed with branches of the inferior mesenteric artery and are inhibitory to colonic muscle.

The midgut receives its parasympathetic innervation from the vagus, via the coeliac and superior mesenteric plexuses, whereas the hindgut receives its parasympathetic innervation from the pelvic splanchnic nerves. The cell bodies of the pelvic splanchnic nerves are located in the second to fourth sacral spinal segments, the sacral parasympathetic nucleus. These parasympathetic fibres enter the inferior hypogastric plexus, where some synapse. From here, some pass directly to the rectum and other pelvic viscera while others ascend by one of two routes: either within the hypogastric nerves to the superior hypogastric plexus to be distributed along branches of the inferior mesenteric artery, or by passing directly through the retroperitoneal tissues to reach the splenic flexure and descending and sigmoid colon. Most preganglionic parasympathetic neurones synapse in intramural plexuses in the gut wall; from here, postganglionic neurones innervate the glands (secretomotor) and muscle (motor) of the large intestine. Parasympathetic stimulation is integral to colonic propulsion and defecation and to relaxation of the internal anal sphincter.

Visceral afferent impulses mediating sensations of distension and spasm from the midgut travel with the vagus nerve while the hindgut is innervated by afferent neurones with cell bodies in the lumbar (mostly L2 and L3) and sacral dorsal root ganglia (mostly S1 and S2) (Brookes et al 2009). These travel alongside autonomic nerves and are often erroneously referred to as 'sympathetic afferents' or 'parasympathetic afferents'. Visceral afferent innervation of intramural colonic blood vessels probably conveys the sensation of colonic distension (Song et al 2009), while visceral afferents in the rectum convey the sensation of rectal filling and are involved in reflex propulsive activity.

MOTOR FUNCTION OF THE LARGE INTESTINE

Healthy colorectal function includes absorption of water and sodium from the bowel lumen, net antegrade propulsion of intestinal contents at an adequate rate, and temporary faecal storage. Colonic motor activity is greatest during the day and increases on waking and after meals. It is characterized by sustained tonic contractions and brief phasic contractions. Phasic contractions are subdivided according to whether or not they propagate along the colon. Non-propagating contractions are associated with segmental mixing of luminal contents, whereas propagated sequences can be retrograde or antegrade. In the normal colon, antegrade activity occurs more frequently than retrograde sequences, which are seen more in the proximal colon. High-amplitude propagated contractions occur in adults 4–6 times per day, originate anywhere in the colon (but mostly proximally), and migrate for a variable distance distally (Narducci et al 1987, Bharucha 2012). They are often associated with defecation or passing flatus. The sigmoid colon exhibits cyclical bursts of motor activity, which may be important in modulating the delivery of faeces to the rectum. A high-pressure zone with unique contractile properties in response to sigmoid and rectal distension or contraction has been demonstrated in the distal sigmoid (the so-called rectosigmoid sphincter; Shafik et al 2003). Rectal motor activity is characterized by motor complexes that can be triggered by propagating contractions from the proximal colon and by the delivery of stool or gas from the sigmoid colon. However, since these frequently propagate retrogradely, it has been suggested that they help to keep the rectum empty, thereby preventing the untimely delivery of colonic contents.

ANAL CANAL

The anal canal begins at the anorectal junction and ends at the anal verge (Figs 66.44–66.46). It is directed posteriorly because the sling-like puborectalis component of levator ani pulls the rectum forwards to

create the anorectal angle. The anal canal lies 2–3 cm anterior and slightly inferior to the tip of the coccyx, opposite the apex of the prostate in males. At the anal verge, the squamous epithelium lining the lower anal canal becomes continuous with the skin of the perineum. The area of pigmentation of skin around the anal verge corresponds approximately to the extent of the external anal sphincter. Identification of the anal verge may be difficult, particularly in males in whom the perineum may 'funnel' upwards into the lower anal canal, but the characteristic puckering of the skin formed by the penetrating fibres of the conjoint longitudinal muscle of the anal canal provides a useful landmark.

The anal canal consists of an inner epithelial lining, a vascular subepithelium, the internal and external anal sphincters, and fibromuscular supporting tissue, as well as dense neuronal networks of autonomic and somatic origin. Functionally, it represents a zone of high pressure. It is between 2 and 5 cm long in adults; the anterior wall is slightly shorter than the posterior. It is usually shorter in females. At rest, it forms an oval or triradiate slit in the anteroposterior plane rather than a truly circular canal. The arrangement of the external anal sphincter and its attachments to the perineal body and coccyx create sites of maximum pressure in the anterior and posterior midline of the canal.

Anteriorly, the middle third of the anal canal is attached by dense connective tissue to the perineal body, which separates it from the membranous urethra in males and from the lower vagina in females. Laterally and posteriorly, the anal canal is surrounded by the loose adipose tissue of the ischio-anal fossae; this arrangement allows expansion of the canal but offers a potential pathway for the spread of perianal sepsis. Posteriorly, the anal canal is attached to the coccyx via the anococcygeal ligament, a midline fibroelastic structure that runs between the posterior aspect of the middle region of the external anal sphincter and the coccyx. The anococcygeal ligament is traditionally regarded as lying just inferior to the midline raphe of levator ani but its relationship to the raphe is more complex (Kinugasa et al 2011).

The ischial spines may be palpated laterally by an examining finger in the upper anal canal. The pudendal nerves pass over the attachment of the sacrospinous ligament at this point and pudendal nerve motor terminal latency may be measured digitally using a modified electrode worn on the examining glove.

LINING OF THE ANAL CANAL

The upper part of the anal canal is lined by reddish columnar epithelium similar to that of the rectum; it contains secretory and absorptive cells with numerous tubular glands or crypts. Distally, the epithelium becomes cuboidal and darker just above the level of the anal valves (Fig. 66.47). The subepithelial tissues are mobile and relatively distensible, and contain submucosal arterial and venous plexuses.

In the mid-anal canal, there are 6–10 vertical mucosal folds, the anal columns. These are better defined in children than in adults. The columns frequently contain a terminal branch of the superior rectal artery and vein, supplemented to a variable degree by middle and inferior rectal vessels (Thomson 1975). Dilated submucosal veins in the upper anal canal form an internal haemorrhoidal venous plexus. Tiny arteriovenous connections to these dilated submucosal veins give the blood within them a higher oxygen tension and therefore a redder colour than normal venous blood. The submucosal vessels are most prominent in the left-lateral, right-posterior and right-anterior quadrants of the wall of the canal (approximately 3, 7 and 9 o'clock when viewed in the lithotomy position); here, the subepithelial tissues are expanded into three 'anal cushions'. Although variable in number and position, the cushions help to seal the anal canal and contribute to the maintenance of continence to flatus and fluid. The anal cushions are important in the pathogenesis of haemorrhoids. The lower ends of the columns form small crescentic folds, called anal valves, between which lie small recesses known as anal sinuses. The anal valves and sinuses together form the scalloped dentate (or pectinate) line, which is firmly anchored to submucosal connective tissue. Anal glands open into small depressions, anal crypts, in the anal valves. The glands are branched and lined by stratified columnar epithelium. Cystic dilations in the glands may extend through the internal anal sphincter and even into the external sphincter (Seow-Cheon and Ho 1994).

The epithelium below the dentate line is smooth and typically shows an abrupt transition to parchment-coloured, non-keratinized, stratified squamous epithelium, which lacks sweat and sebaceous glands and hair follicles but contains numerous somatic sensory nerve endings conveying sensations of touch, pain and temperature. It extends down to the intersphincteric groove, a palpable depression at the lower border of the internal sphincter. Below the intersphincteric groove, the anal canal

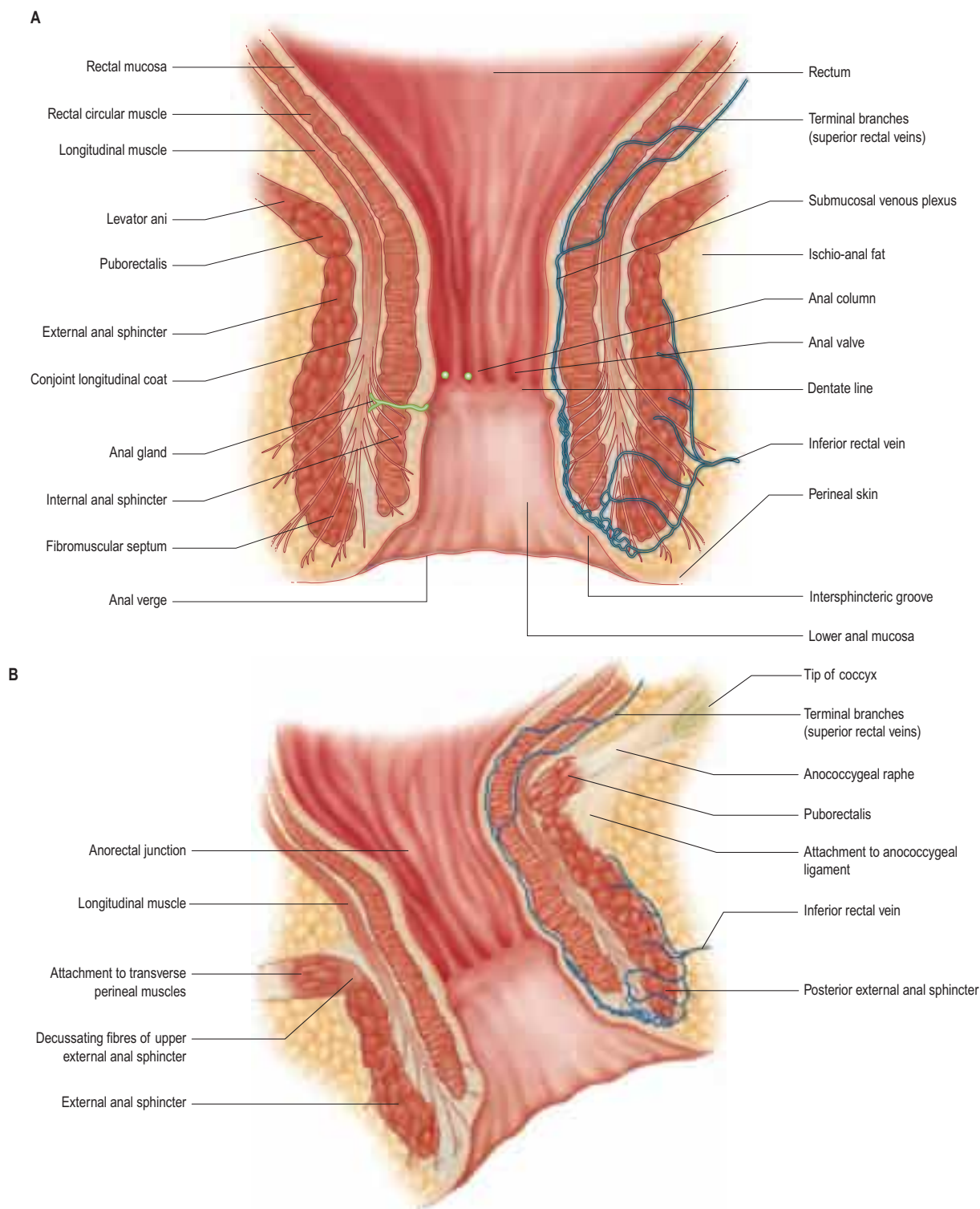


Fig. 66.44 **A**, A coronal section through the anal canal; glandular and vascular structures are shown unilaterally for clarity. **B**, A sagittal section through the anal canal; the anal canal is angled posteriorly and so anterior sphincteric structures appear to be lower than posterior structures. The glandular, vascular and fibromuscular structures have been omitted for clarity.

is lined by hair-bearing keratinizing, stratified squamous epithelium, which is continuous with the perianal skin. Dilated veins in the subepithelial tissue below the dentate line contribute to an external haemorrhoidal venous plexus (see Fig. 66.44); this can be seen on clinical examination as a circumferential ring at the anal margin if the subject is asked to bear down.

The junction between the columnar epithelium above and the squamous epithelium below is referred to as the anal transition zone (ATZ). It is variable in length and position but often extends about 1 cm proximal to the dentate line (Fenger 1987). Nerve endings, including thermoreceptors, exist in the submucosa around the upper anal transition zone; they probably play a role in continence by providing a highly specialized 'sampling' mechanism to identify the contents of the lower rectum when the upper anal canal relaxes (Duthie and Gairns 1960).

The well-defined muscularis mucosae of the rectum continues into the upper canal. Some fibres from the conjoint longitudinal muscle of the anal canal traverse the internal anal sphincter and turn cranially to merge with the muscularis mucosae, forming the submucosal muscle of the anal canal (also called the musculus submucosae ani, or musculus canalis ani, or Treitz's muscle). Attachments of the longitudinal fibres to the submucosal muscle of the anal canal are particularly dense at the dentate line (the mucosal suspensory ligament (of Parks)), causing the mucosa to be tethered more firmly at this level and approximately marking the watershed between the portal and systemic venous drainage of the haemorrhoidal plexuses. The attachments of fibres from the longitudinal muscle layer, the muscularis mucosae of the anal canal, also create 'spaces' within the submucosa: a proximal submucous space, a marginal space extending from the dentate line to the intersphincteric groove, and a perianal space distally.



Fig. 66.45 **A**, A mid-coronal MRI endocoil image of the anal canal. **B**, An anterior coronal MRI endocoil section in a woman showing the transverse perineal (TP) joining the external anal sphincter anteriorly (between arrows). Other abbreviations: EAS, external anal sphincter; IAS, internal anal sphincter; PR, puborectalis.

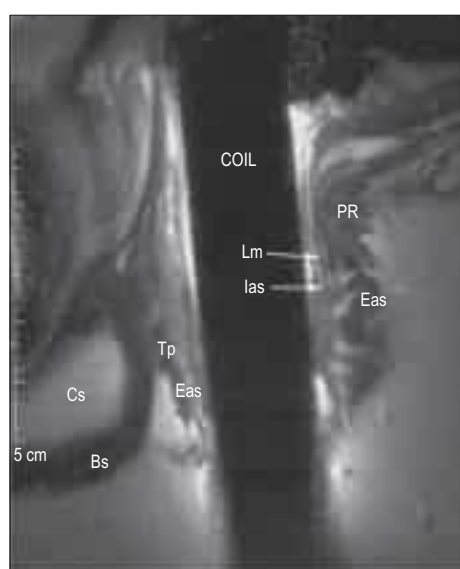


Fig. 66.46 An MRI endocoil mid-sagittal view of the anal canal in a man. Abbreviations: Bs, bulbospongiosus; Cs, corpus spongiosum; Eas, external anal sphincter; Ias, internal anal sphincter; Lm, longitudinal muscle; PR, puborectalis; Tp, transverse perineal.



Fig. 66.47 The endoscopic appearance of the anorectal junction viewed internally from above. The endoscope is visible in the upper left corner of the image.

MUSCLES OF THE ANAL CANAL

The anal canal is surrounded by internal and external anal sphincters, separated by the conjoint longitudinal muscle layer, the whole arrangement being referred to as the anal sphincter complex (Figs 66.48–66.49) (Al-Ali et al 2009). The anatomy of this region has been clarified in recent years using endoanal sonography and MRI.

Internal anal sphincter

The internal anal sphincter is the specialized, white, thickened terminal part of the inner circular muscle of the large intestine. It is thicker in males and in patients with chronic constipation. The muscle fibre arrangement is not truly circular, but rather a tight spiral that shortens and widens with relaxation. It commences at the anorectal junction and ends above the anal verge; its lower border is palpable at the intersphincteric groove, which corresponds to the proximal limit of the subcutaneous part of the external anal sphincter. It is traversed by fibres passing medially from the conjoint longitudinal muscle into the submucosa.

At rest, it is tonically contracted but it relaxes as a consequence of reflex activity, predominantly during defecation. Transient relaxation of the upper internal anal sphincter occurs in response to rectal distension (the recto-anal inhibitory reflex) and postprandial rectal contractions (the sampling reflex). Relaxation allows the passage of distal rectal contents into the upper anal canal, enabling a conscious or subconscious perception of their physical nature; this is accompanied by sustained contraction of the distal internal anal sphincter and contraction of the external anal sphincter to maintain continence. The recto-anal inhibitory reflex is primarily mediated by the enteric nervous system, although spinal pathways may have a modulatory role. It is absent in patients with Hirschsprung's disease.

Vascular supply and innervation

The internal anal sphincter is supplied by terminal branches of the superior and inferior rectal vessels, and innervated extrinsically by autonomic nerves. Sympathetic fibres originate in the upper two lumbar spinal segments, and parasympathetic fibres originate in the second to fourth sacral spinal segments, both being distributed via the inferior hypogastric plexus. Stimulation of cholinergic muscarinic receptors (parasympathetic) causes internal anal sphincter relaxation and longitudinal anal muscle contraction, while α -adrenergic receptor stimulation (sympathetic) causes contraction of both the internal anal sphincter and longitudinal muscle. Activation of nitrenergic nerves also mediates internal anal sphincter relaxation and is the rationale for using topical nitroglycerine and other drugs that promote nitric oxide release in the treatment of pathological conditions associated with increased resting anal tone.

Conjoint longitudinal muscle

The conjoint longitudinal muscle of the anal canal is a direct continuation of the outer longitudinal smooth muscle of the rectum, descending between internal and external anal sphincters, and augmented in its upper part by striated muscle fibres from the medial aspect of levator ani (Lunniss and Phillips 1992). The muscle is particularly prominent in the fetus, where it is actually thicker than the internal anal sphincter. With advancing age, there is gradual replacement of muscle by connective tissue, such that the layer becomes thin in the elderly and few muscle fibres are seen in its distal part. As it passes down the anal canal, muscle fibres peel off in three directions: internally, through the internal anal sphincter to reach the anal submucosa; inferiorly, through the striated muscle of the lower subcutaneous part of the external anal sphincter to insert into the perianal skin (some of these fibres encircling the anal orifice); and outwards, through the upper part of the external

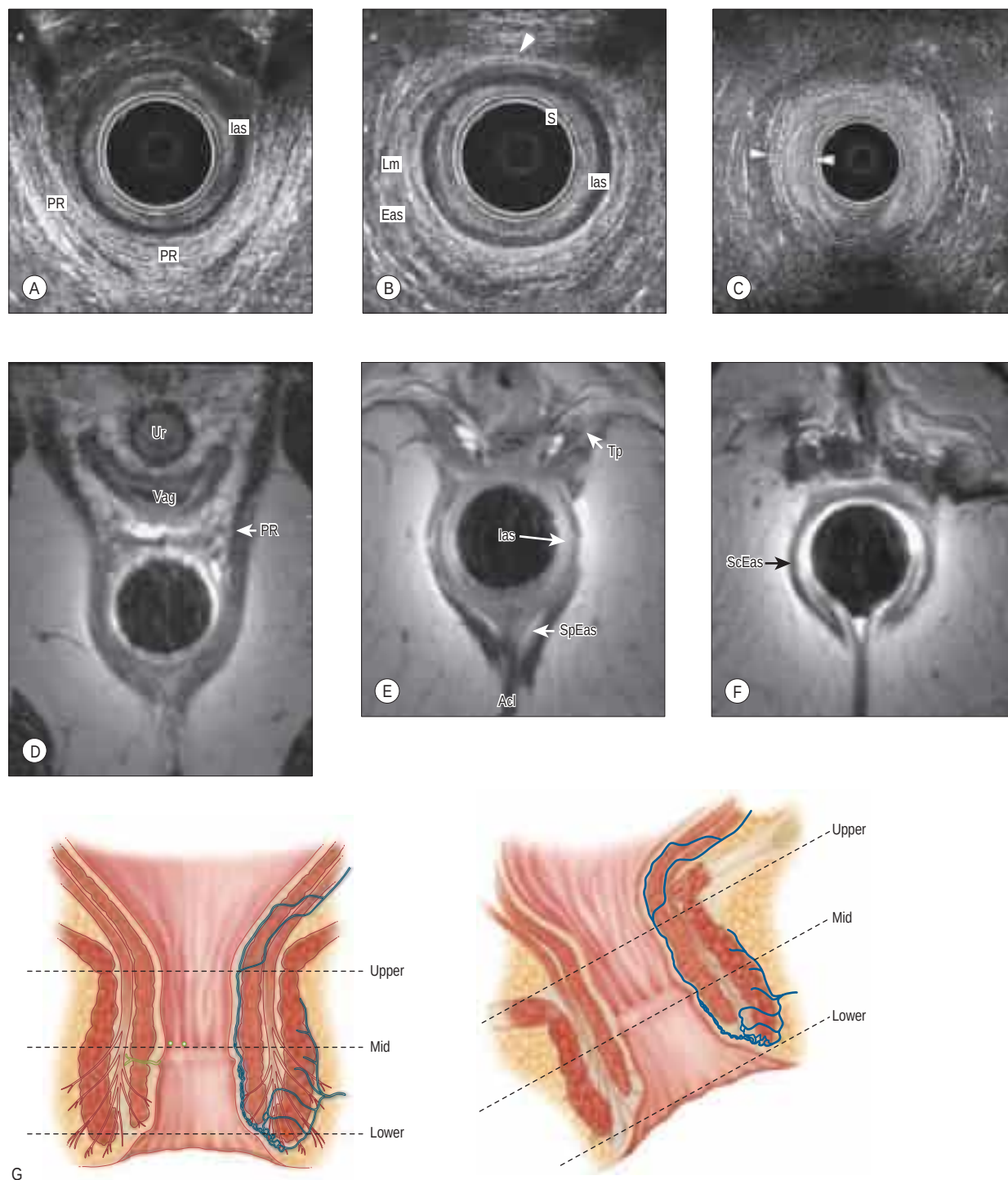


Fig. 66.48 A–C, Axial views of the anal canal at three levels on endoanal ultrasound in a woman. The endoanal ultrasound probe is the central black structure. **A**, The upper anal canal. The 'U' shape of puborectalis is visible. **B**, The middle anal canal. The external anal sphincter is now a complete ring anteriorly (arrowhead). **C**, The lower anal canal. Below the termination of the internal anal sphincter, the longitudinal layer extends through the subcutaneous external anal sphincter (between arrowheads). **D–F**, An MRI of the anal canal. **D**, At upper anal canal level, the sling of puborectalis extends anteriorly to the pubic bones. **E**, At mid anal canal level, the transverse perineii fuse into the external anal sphincter anteriorly. The superficial (middle) external anal sphincter is attached either side of the anococcygeal ligament. **F**, The low anal canal level, below the internal anal sphincter. **G**, A key for levels of the anal canal. Abbreviations: Acl, anococcygeal ligament; Eas, external anal sphincter; las, internal anal sphincter; Lm, longitudinal muscle; PR, puborectalis; S, subepithelial tissues; ScEas, subcutaneous (lower) part of the external anal sphincter; SpEas, superficial (middle) external anal sphincter; Tp, transverse perineii; Ur, urethra; Vag, vagina.

anal sphincter (see Fig. 66.44). The lowermost fibres create a honeycomb arrangement in the subcutaneous fat and separate a superficial perianal space from the deeper ischio-anal fossa. Additional ramifications of this muscle beyond the external anal sphincter have been described, emphasizing its central role in anorectal stability (Courtney 1950).

The conjoint longitudinal muscle is innervated by autonomic nerves that share their origin with those supplying the internal anal sphincter (see above). Its contraction during defecation shortens and widens the anal canal, and everts the anal orifice. Degeneration of the muscle and its gradual replacement by connective tissue occurs with age (Haas et al

1984) and is central to the development of haemorrhoids. Its extensions provide pathways for the spread and containment of infection.

External anal sphincter

The external anal sphincter forms the bulk of the anal sphincter complex. It is an oval tube of striated muscle composed mostly of type I slow twitch muscle fibres adapted for prolonged contraction. The length and thickness of the external anal sphincter is less in females (Rociu et al 2000). The historical concept of the muscle having three parts (deep, superficial and subcutaneous) is no longer valid, but upper (deep) and lower (superficial or subcutaneous) parts are described by

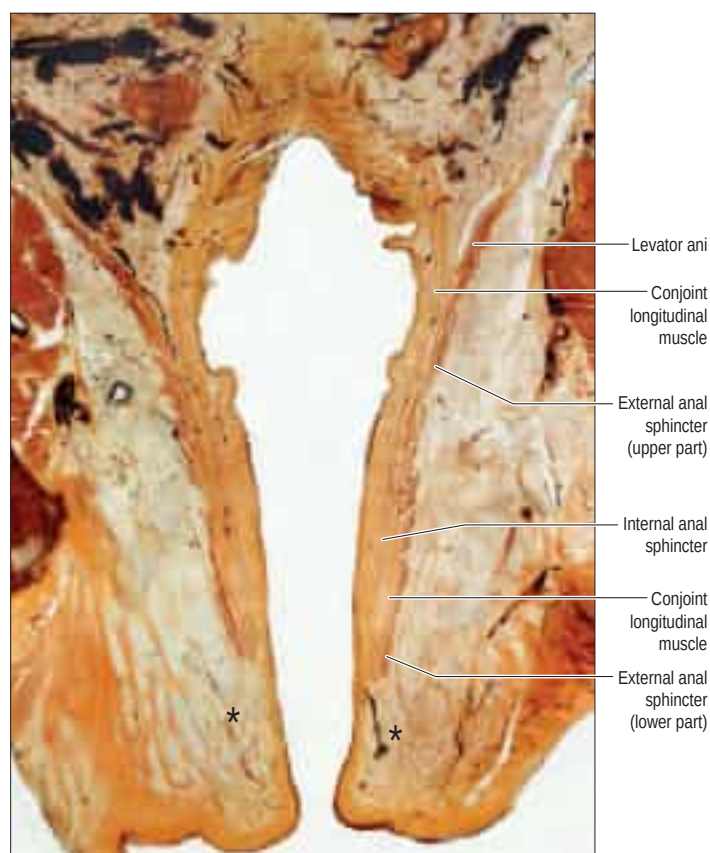


Fig. 66.49 A transilluminated 2.5 mm thick coronal section of a male pelvis processed for epoxy resin E12 sheet plastination, showing the entire length of the anal canal and its muscular layers: namely, the external anal sphincter, conjoint longitudinal muscle and internal anal sphincter. Note the continuity of the external anal sphincter with levator ani superiorly, consisting of two parts: the upper part and the lower part. The upper part of the conjoint longitudinal muscle is continuous with the longitudinal muscular layer of the rectum and, inferiorly, its fibres traverse the lower part of the external anal sphincter to blend with or form fibro-fatty, honeycomb-like compartments around the perianal region (asterisks). The internal anal sphincter ends above the lower part of the external anal sphincter. (Courtesy of Dr Saad Al-Ali, with permission from Al-Ali S, Blyth P, Beatty S, Duang A, Parry B, Bissett IP. (2009) Correlation between gross anatomical topography, sectional sheet plastination, microscopic anatomy and endoanal sonography of the anal sphincter complex in human males. *J. Anat.* 215; 212–220.)

some authors. The upper part surrounds the internal anal sphincter while the lowermost part encircles the anal canal inferior to the internal anal sphincter (Fritsch et al 2002, Al-Ali et al 2009).

The upper part of the external anal sphincter is attached to the anococcygeal ligament posteriorly and to the perineal body anteriorly; some muscle fibres on each side of the sphincter decussate to form a commissure in the anterior and posterior midline. The uppermost fibres blend with the lower medial fibres of puborectalis and attach to the anococcygeal raphe posteriorly and the transverse perineal muscles anteriorly. The lower part of the external anal sphincter extends below the internal anal sphincter and is traversed by the terminal fibres of the conjoint longitudinal muscle (see above). Anteriorly, it is attached to bulbospongiosus and bulbocavernosus (Fig. 66.50) (Shafik et al 2007). Like the levator ani and internal anal sphincter, the external anal sphincter is tonically contracted at rest (the postural reflex).

Vascular supply and innervation

The external anal sphincter is supplied by the inferior rectal vessels, with a small contribution from the median sacral artery. It is innervated bilaterally by the inferior rectal branch of the pudendal nerve (containing contributions from the second, third and fourth sacral spinal nerves). The pudendal nerve also carries afferent fibres from the lining of the anal canal and perianal skin. The upper external anal sphincter may also receive motor fibres from the nerve to levator ani (ventral rami of predominantly the third and fourth sacral spinal nerves).

Intersphincteric space and anal glands

The intersphincteric 'space' is a potential space between the conjoint longitudinal muscle layer and the external anal sphincter. It can be

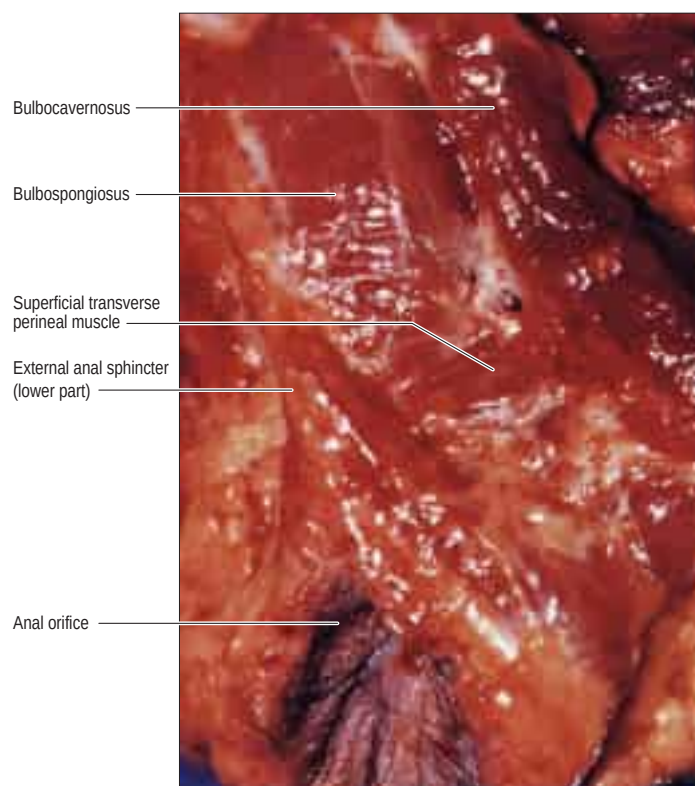


Fig. 66.50 A dissection of the male external anal sphincter in a fresh cadaver, viewed from below. (Courtesy of Dr Robert P Myers.)

entered surgically to provide access in a variety of operations (e.g. intersphincteric excision of the rectum and intersphincteric approaches to fistulae). Within the space lie the intersphincteric anal glands, the source of most anal fistulae (Parks 1961). There is an average of twelve intersphincteric anal glands within the adult anal canal, evenly distributed around the circumference. Their function is unknown, but they secrete mucin (different in composition from that secreted by rectal mucosa) and they communicate with the anal lumen via ducts (lined by epithelium similar to that of the anal transition zone), which cross the internal anal sphincter to open at the level of the anal valves immediately above the dentate line (Seow-Choen and Ho 1994). Retrograde passage of bacteria from the anal canal to the gland is understood to cause infection; inflammatory occlusion of the duct prevents spontaneous drainage back into the lumen of the anal canal. Sepsis may then spread along a variety of routes to cause abscesses and fistulae within various spaces (Parks et al 1976).

VASCULAR SUPPLY AND LYMPHATIC DRAINAGE OF THE ANUS

Arteries

The anal canal is supplied by terminal branches of the superior rectal artery and the inferior rectal branch of the internal pudendal artery, together with a small contribution from the median sacral artery. The arterial supply to the epithelium of the lower anal canal in the midline, particularly posteriorly, is relatively deficient (Klosterhalfen et al 1989); this is further diminished if the internal anal sphincter is hypertonic (Schouten et al 1996). The epithelium is more firmly tethered to underlying structures in the midline, which may also be a focal point of pressure in the anal canal. Collectively, these factors are thought to predispose to the occurrence of acute and chronic anal fissures, which are most commonly found in the midline, especially posteriorly.

Veins

The venous drainage of the anal canal parallels the arterial supply. The upper canal is drained predominantly by the superior rectal veins, tributaries of the inferior mesenteric vein and the portomesenteric venous system; some blood returns to the internal iliac veins via the middle rectal veins. The lower part of the anal canal, including the external haemorrhoidal venous plexus, drains via the inferior rectal branches of the internal pudendal vein into the internal iliac vein.

Haemorrhoids

Haemorrhoids develop when the supporting connective tissues of the anal cushions degenerate, causing downward displacement of the cushions and abnormal venous dilation (Lohsiriwat 2012). They are associated with laxity of the anal canal submucosa and enlargement of the terminal branches of the superior rectal artery supplying the anal cushions.

Lymphatic drainage

Lymph from the upper anal canal drains cranially into the submucosal and intramural lymphatics of the rectum. Lymphatics around puborectalis drain to internal iliac lymph nodes. Lymph from the lower anal canal and external sphincter drains to inguinal lymph nodes. This pattern of drainage is particularly important when considering lymphatic spread of malignant tumours from the lower rectum and anal canal. Blockage of lymph drainage along normal routes can lead to unusual patterns of dissemination.

DEFECATION

Defecation is the act of voiding stool from the anus and involves the coordinated function of the colon and rectum, pelvic floor and anal sphincter. It is influenced by multiple factors, including colonic transit, stool volume and consistency, posture, age, gender, psychology and behaviour, culture and lifestyle (Palit et al 2012). Social behaviour entrains defecation to occur at a time of convenience rather than simply in response to a defecatory urge. The urge to defecate is preceded by high-amplitude propagated contractions in the colon, which result in the antegrade propulsion of intraluminal contents to the rectum; these contractions increase in frequency and intensity just before expulsion. The response to rectal filling involves intact rectal afferent nerves, reflex rectal contractions and normal rectal wall biomechanics. The rectum acts as a reservoir and is able to relax to accommodate faecal material and gas (adaptive relaxation). Periodic sampling within the upper anal canal helps to determine the nature of the rectal content. If the defecatory urge is inconvenient, puborectalis

and the external anal sphincter, which are tonically contracted at rest, can be further contracted voluntarily to move stool back up from the distal rectum. If the defecatory urge is associated with a conscious decision to evacuate, distal progression of colonic high-amplitude propagating complexes, rectal contractions, raised intra-abdominal pressure from voluntary straining (the efficiency of which is influenced by posture), relaxation of puborectalis (which straightens the anorectal angle) and the anal sphincters all combine to enable defecation. Relaxation of the internal anal sphincter is a reflex response to rectal distension, whereas relaxation of the external anal sphincter is voluntary. During defecation, contraction of the conjoint longitudinal muscle shortens and opens the anal canal and flattens the anal cushions. Both rectal and a variable quantity of colonic contents are evacuated. After defecation, the external anal sphincter contracts (the closing reflex), the internal anal sphincter gradually recovers its resting tone, and the postural reflex is reactivated. The conjoint longitudinal muscle relaxes, permitting the anal canal to elongate and the anal cushions to re-expand.

Problems with defecation may result from impaired colonic transit or from a variety of functional or mechanical rectal disorders (Lunniss et al 2009). Functional causes include inadequate or ineffective expulsive effort, paradoxical contraction of the pelvic floor and external anal sphincter (dyssynergia), and lack of normal sensation (rectal hyposensitivity). Mechanical causes include impedance to evacuation when straining (rectal intussusception, obstructing masses or anal stenosis), misdirection of force vectors during straining (rectocele), or dissipation of evacuatory forces (ballooning perineum).

Anal continence and incontinence

As long as anal pressure exceeds rectal pressure, faecal continence will be maintained. Anal resting tone is derived mainly from sympathetically mediated tonic contraction of the internal anal sphincter, augmented by contraction of the external anal sphincter. The anal cushions contribute by providing an effective seal. Anal continence involves central, spinal and peripheral pathways, somatic, autonomic and enteric nerves, intact sacral sensorimotor reflexes, and structurally and functionally intact smooth and striated musculature.

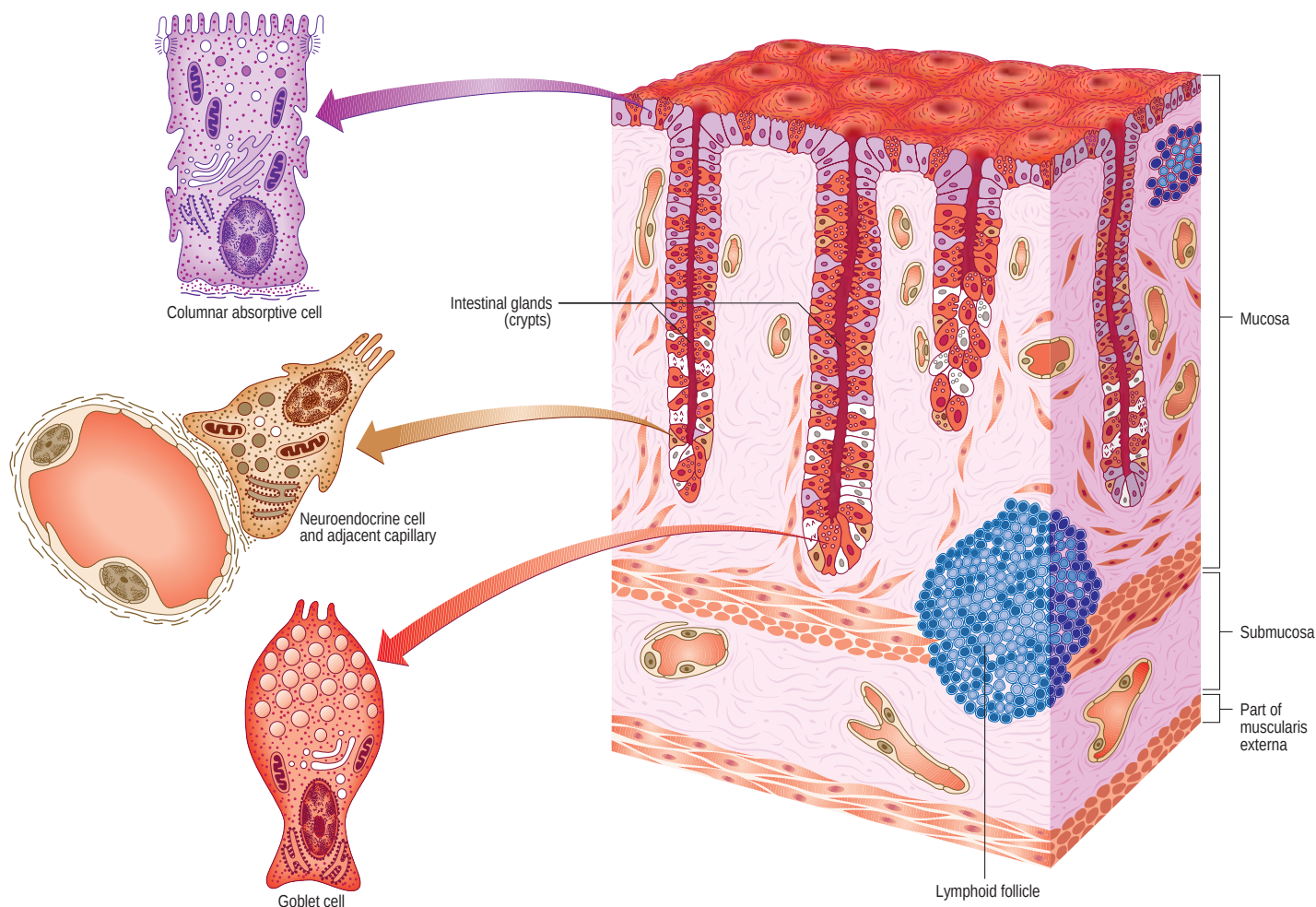


Fig. 66.51 The microstructure of the colonic wall and its epithelial cells. Note the aggregations of lymphocytes (blue) and undifferentiated epithelial cells (white).

Sphincteric causes of faecal incontinence may be structural (disruption or atrophy of part of the sphincter musculature) or neuropathic (damage to the nerve supply to the sphincters), or a combination of both. The most common causes of sphincter disruption are obstetric injury, anal surgery (for haemorrhoids, fistula or fissure) and trauma. Pudendal neuropathy is most commonly associated with childbirth (prolonged second stage) and chronic straining.

Although the anus constitutes the final barrier to faecal incontinence, suprasphincteric factors are also essential in maintaining faecal continence, particularly the rate at which the stool is delivered to the rectum, rectal sensation and compliance, and adaptive relaxation. Thus, even in the presence of structurally and functionally intact sphincter mechanisms, faecal incontinence may occur: for instance, in diarrhoeal states or conditions where there is loss of rectal reservoir function (Scott and Lunniss 2007). Conversely, chronic rectal distension with retained faeces may result in passive (overflow) leakage, possibly as a result of a chronically relaxed internal anal sphincter consequent on a persistently activated recto-anal inhibitory reflex, together with blunted rectal sensation causing diminished conscious contraction of the external anal sphincter.

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MICROSTRUCTURE OF THE LARGE INTESTINE



Available with the *Gray's Anatomy* e-book

The layers of tissue in the large intestinal wall (Fig. 66.51) resemble those in the small intestine (Ch. 65), except that villi and circular folds are absent and the glands (crypts) are longer.



Bonus e-book image

Fig. 66.29 A CT scan (coronal reformat) showing both right colonic and sigmoid diverticulosis.

Mucosa

The mucosa is pale, smooth and, in the colon, raised into numerous crescent-shaped folds between the sacculi. It is thicker, darker, more vascular, and more loosely attached to the submucosa in the rectum.

Epithelium

The luminal surface of all but the anorectal junction is lined by columnar cells, mucous (goblet) cells and occasional microfold (M) cells that are restricted to the epithelium overlying lymphoid follicles.

Columnar (absorptive) cells Columnar (absorptive) cells are the most numerous of the epithelial cell types. Although there is some variation in their structure, they all bear apical microvilli which are shorter and less regular than those on enterocytes in the small intestine. Typical junctional complexes around their apices limit extracellular diffusion from the lumen across the intestine wall.

Mucous (goblet) cells Mucous cells have a similar structure to those of the small intestine but they are more numerous. They are outnumbered by absorptive cells for most of the length of the colon but they are equally frequent towards the rectum, where their numbers increase.

Microfold (M) cells Microfold cells are similar to those of the small intestine.

Intestinal glands (crypts)

Intestinal crypts are narrow, perpendicular tubular glands that are longer, more numerous and closer together than those of the small intestine. Their orifices lend a cribriform appearance to the mucosa in surface view. The glands are lined by low columnar epithelial cells, mainly goblet cells, augmented by columnar absorptive cells and neuroendocrine cells. The latter are situated mainly at the bases of the glands, and secrete basally into the lamina propria. In general, the glands lack Paneth cells but some may be present in the caecum. Stem cells located at or near the bases of the intestinal glands (crypts) are the source of the other epithelial cell types in the large intestine. They provide cells that migrate towards the luminal surface of the intestine; their progeny differentiate, undergo apoptosis and are shed after approximately 5 days.

Lamina propria

The lamina propria is composed of connective tissue that supports the epithelium, forming a specialized pericryptal myofibroblast sheath around each intestinal gland. Solitary lymphoid follicles within the lamina propria, similar to those of the small intestine, are most abundant in the caecum, appendix and rectum, but are also present scattered along the rest of the large intestine; efferent lymphatic vessels originate within them. Lymphatic vessels are absent from the intercryptal lamina propria.

Muscularis mucosae

The muscularis mucosae of the large intestine is essentially similar to that of the small intestine.

Submucosa

The submucosa of the large intestine is similar to that of the small intestine.

Muscularis externa

The muscularis externa has outer longitudinal and inner circular layers of smooth muscle. The longitudinal fibres form a continuous layer that is aggregated into macroscopically visible longitudinal bands or taeniae coli (see Figs 66.3 and 66.13). Between the taeniae coli, the longitudinal layer is much thinner, less than half the thickness of the circular layer. The circular fibres constitute a thin layer over the caecum and colon, and a thicker layer in the walls of the rectum; they form the internal anal sphincter in the anal canal. Interchange of fascicles between circular and longitudinal layers occurs, especially near the taeniae coli. Deviation of longitudinal fibres from the taeniae coli to the circular layer may, in some instances, explain the haustrations of the colon.

Serosa

The serosa or visceral peritoneum is variable in extent. Small, fat-filled appendices epiploicae are most numerous on the sigmoid and transverse colon but generally absent from the rectum.

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